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Depletion of SMN Protein in Mesenchymal Progenitors Impairs the Development of Bone and Neuromuscular Junction in Spinal Muscular Atrophy

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Abstract

Spinal Muscular Atrophy (SMA) is a neuromuscular disorder characterized by the deficiency of the survival motor neuron (SMN) protein, which leads to motor neuron dysfunction and muscle atrophy. In addition to the requirement for SMN in motor neurons, recent studies suggest that SMN deficiency in peripheral tissues plays a key role in the pathogenesis of SMA. Using limb mesenchymal progenitor cells (MPCs)-specific SMN-depleted mouse models, we reveal that SMN reduction in chondrocytes and fibro-adipogenic progenitors (FAPs) derived from limb MPCs causes defects in the development of bone and neuromuscular junction (NMJ), respectively. We showed that impaired growth plate homeostasis, which causes skeletal growth defects in SMA, is due to reduced IGF signaling from chondrocytes rather than the liver. Furthermore, the reduction of SMN in FAPs resulted in abnormal NMJ maturation, altered release of neurotransmitters, and NMJ morphological defects. Transplantation of healthy FAPs rescued the morphological deterioration. Our findings highlight the significance of mesenchymal SMN in neuromusculoskeletal pathogenesis in SMA and provide insights into potential therapeutic strategies targeting mesenchymal cells for the treatment of SMA.

eLife assessment

This **important** work by Hann et al. advances our understanding of the role of the survival motor neuron (SMN) protein in coordinating pathogenesis of the spinal muscular atrophy (SMA). The authors addressed many concerns raised by the reviewers, providing **convincing** evidence in terms of skeletal analyses not being able to satisfactorily elucidate SMN regulation of bone development.

Introduction

The survival motor neuron (SMN) protein is a crucial component of the spliceosome complex and is essential for the proper function of all cell types (Mercuri et al., 2022). Deficiency in SMN protein disrupts the formation of spliceosome complexes, ultimately causing splicing defects in multiple genes. Mutations in the *SMN1* gene, which encodes the SMN protein, give rise to the neuromuscular disorder Spinal Muscular Atrophy (SMA). SMA is characterized by neuromuscular junctions (NMJ) disruption, muscular atrophy, and alpha motor neuron loss (Mercuri et al., 2022; Burghes et al., 2009). The severity of the disease in humans correlates with the copy number of *SMN2*, which is a paralog of *SMN1* in humans. *SMN2* primarily produces less functional exon 7-deleted SMN protein and rarely generates a limited quantity of functional full-length SMN protein via alternative splicing. SMA patients are classified into types 0 through 4 based on the severity and timing of disease onset, predominantly determined by the number of copies of the *SMN2* gene they possess. More than 50% of SMA patients are categorized as type 1, characterized by muscle defects with proximal muscle atrophy during infancy, eventually resulting in death within a few years (Mercuri et al., 2022).

Previous studies suggest that the onset of SMA is mainly attributed to SMN loss in motor neurons (Monani et al., 2000; Burghes et al., 2009). Nevertheless, motor neuron-specific SMN deficiency in the SMA mouse model exhibits relatively mild phenotypes compared to whole-body SMA mouse models (Park et al., 2010; McGovern et al., 2015). Furthermore, restoring SMN to motor neurons in SMA mouse models result in only partial rescue in lifespan and neuromuscular defects (Passini et al., 2010; Martinez et al., 2012; McGovern et al., 2015; Besse et al., 2020). Systemic administration of antisense oligonucleotide (ASO), which corrects *SMN2* splicing to restore SMN expression, significantly prolongs survival compared to CNS administration (Hua et al., 2011). In the mouse treated with a systemically delivered ASO, blocking the effect of ASO in the CNS by a complementary decoy did not have any detrimental effect on survival, motor function, or NMJ integrity (Hua et al., 2015). These studies suggest that peripheral SMN plays a crucial part in SMA pathology. Overall, investigating the impacts of SMN depletion in peripheral tissues is critical for alleviating neuromuscular impairments and increasing life expectancy in SMA.

In SMA patients, bone growth retardation has been observed (De-Amicis et al., 2021; Kipoglu et al., 2022; Hensel et al., 2020). Studies using whole-body SMA mouse models have revealed that this is caused by diminished growth plate chondrocyte density and endochondral ossification defects, independent of muscle atrophy (Hensel et al., 2020). However, it is still unclear whether these defects result from the SMN ablation in chondrocytes, or from a decline in liver-derived IGF in SMA patients and mice (Hua et al., 2011; Yesbek-Kaymaz et al., 2016). In severe SMA mice, the serum levels of IGF decreased by approximately 60% or became undetected (Hua et al., 2011; Murdocca et al., 2012). The decreased serum IGF levels were attributed to decreased expression of liver genes, including *Igf1*, IGF binding, and ternary complex protein gene *Igfals* and *Igfbp3*. The previous studies suggest that the liver is the primary origin of systemic IGF, as demonstrated by the liver-specific deletion of *Igf1* and the knockout of *Igfals*, which is mostly expressed in the liver (Yakar et al., 1999; Yakar et al., 2002). The double KO mice exhibited a 90% decrease in serum IGF levels and displayed a phenotype of shortened femur length and growth plates. It is thus possible that the decrease in serum IGF levels, resulting from reduced liver IGF pathway genes in SMA mice, has also played a role in the observed bone growth defect.

Mesenchymal progenitor cells (MPCs) derived from the lateral plate mesoderm (LPM) differentiate into various types of limb mesenchymal cells, including bone, cartilage, and intramuscular mesenchymal cells like fibro-adipogenic progenitors (FAPs) (Nassari et al., 2017). Recent studies have revealed the role of FAPs in skeletal muscle homeostasis, as reducing the number of FAPs resulted in diminished muscle regeneration capacity, long-term muscle atrophy, and NMJ

denervation (Wosczyzna et al., 2019 [↗](#); Uezumi et al., 2021 [↗](#)). Our latest research demonstrated that FAPs-specific deficiency of *Bap1*, one of the deubiquitinases, leads to NMJ defects (Kim et al., 2022 [↗](#)). These recent findings raise the possibility that FAPs may have a specific role in the pathogenesis of neuromuscular diseases such as SMA. However, it has not been studied if depletion of SMN in FAPs can lead to SMA-like neuromuscular pathology.

In this study, we crossed a limb MPCs-specific Cre mouse with a floxed *Smn* exon7 mouse carrying multiple copies of the human *SMN2* gene, allowing us to examine the impact of mesenchymal SMN reduction on SMA pathogenesis. Our findings indicate that the SMN reduction in FAPs, similar to the extent of severe SMA, causes altered NMJ development like SMA. Additionally, in growth plate chondrocytes, it leads to skeletal growth abnormalities due to defects in autocrine and paracrine signaling of IGF.

Results

Bone growth restriction due to growth plate cell-autonomous defects caused by MPCs-specific SMN depletion

To investigate the effects of SMN reduction within MPCs in SMA pathogenesis, we crossed *Smn*^{f7/f7} mice, which possess loxP sites flanking exon 7 of the *Smn* gene (Frugier et al., 2000 [↗](#)), with *Prrx1*^{Cre} mice. This produced *Smn*^{ΔMPC} mice (*Prrx1*^{Cre}; *Smn*^{f7/f7}) that lacked the *Smn* gene specifically in *Prrx1*^{Cre}-expressed limb MPCs that give rise to bone, cartilage, and FAPs (Logan et al., 2002; Leinroth et al., 2022 [↗](#)). To ascertain whether mutant mice carrying the *SMN2* gene, like SMA patients, present pathological phenotypes, we additionally generated *SMN2* 2-copy *Smn*^{ΔMPC} (*Prrx1*^{Cre}; *Smn*^{f7/f7}; *SMN2*^{+/+}) and *SMN2* 1-copy *Smn*^{ΔMPC} (*Prrx1*^{Cre}; *Smn*^{f7/f7}; *SMN2*^{+/-}) mice. Control littermates that lacked *Prrx1*^{Cre} were used as controls for comparison.

SMN2 0-copy *Smn*^{ΔMPC} mice died within 24 hours after birth. Regions where limbs should have formed at E18.5 only had rudimentary limb structures (**Figure 1—figure supplement 1A** [↗](#)). Furthermore, since the upper head bone did not cover the brain, it was directly attached to the skin and protruded. These observations can be attributed to *Prrx1*^{Cre}-mediated SMN deletion in the LPM-derived limb MPCs and the craniofacial mesenchyme, which is accountable for the formation of calvarial bone (Wilk et al., 2017 [↗](#)). To investigate whether the lack of SMN proteins in MPCs is responsible for bone development abnormalities, we performed alcian blue and alizarin red staining on E18.5 *SMN2* 0-copy *Smn*^{ΔMPC} mutants to analyze the structure of bones and cartilage. The appendages displayed restricted bone and cartilage formations, with scarcely discernible femur and tibia (**Figure 1—figure supplement 1B** [↗](#)–**C** [↗](#)). In the cranial region, there was an absence of both cartilage and bone at the location of the calvarial bone, with the parietal bone entirely missing and partially absent frontal bone (**Figure 1—figure supplement 1D** [↗](#)). Additionally, the sternum, which is one of the bones originating from the LPM (Sheng et al., 2015 [↗](#)), was shorter than the control (**Figure 1—figure supplement 1E** [↗](#)).

The *SMN2* 2-copy *Smn*^{ΔMPC} mice, carrying two homologous *SMN2* genes, did not show any discernible differences from the *Prrx1*^{Cre}-negative control littermates into adulthood. However, the *SMN2* 1-copy *Smn*^{ΔMPC} mice exhibited reduced body size and shorter limb length compared to the *SMN2* 1-copy control (*Smn*^{f7/f7}; *SMN2*^{+/-}). To assess postnatal bone growth defects observed in SMA patients and mouse models, we conducted micro-CT analysis on femurs obtained from postnatal day 14 (P14) *SMN2* 1-copy *Smn*^{ΔMPC} and *SMN2* 2-copy *Smn*^{ΔMPC} mice (**Figure 1A** [↗](#)). The 3D reconstruction image showed that the *SMN2* 1-copy mutant femur was smaller than the WT control and *SMN2* 2-copy mutant, and secondary ossification center is denied. The longitudinal virtual section view displayed reduced trabecular bone in the *SMN2* 1-copy mutant femur. CT analysis data showed that *SMN2* 1-copy mutants exhibited reduced femur diaphysis length, diameter, and trabecular bone volume compared to the control group, indicating growth plate-

dependent endochondral ossification defects (**Figure 1B–D**). We then examined femoral bone thickness and diaphyseal bone mineral density (BMD) to determine whether mineralization was normal after bone formation. The thickness of the bone in the mutants did not differ significantly from the control, suggesting that bone mineralization was intact (**Figure 1E** and **Figure 1—figure supplement 2A**). Unexpectedly, BMD slightly increased in SMN2 1-copy mutants compared to the control group (**Figure 1—figure supplement 2B**). To assess the impact of osteoclasts and osteoblasts on diaphysis cortical bone mineralization, we utilized *Itgb3* immunofluorescence as the osteoclast marker and toluidine blue staining for imaging bone-attached osteoblasts (Romeo et al., 2019; Colaianni et al., 2015). Osteoclast and osteoblast density did not significantly differ between the SMN2 1-copy mutant and the control (**Figure 1—figure supplement 2C–E**). The higher BMD may be attributed to greater mechanical stress caused by the shorter femur supporting the weight of the body, consistent with prior research indicating that elevated mechanical force leads to higher BMD in the femur (Hoxha et al., 2014; Ike et al., 2015). Nevertheless, the decrease in bone growth without apparent deterioration in bone mineralization of the femur of SMN2 1-copy *Smn*^{ΔMPC} mutants is consistent with findings from the whole-body SMA mouse model (Hensel et al., 2020). Collectively, these results suggest that mice carrying low copies of *SMN2*, with the *SMN* gene specifically deleted in MPCs, exhibit bone growth abnormalities, while osteoblast and osteoclast populations remain unaffected.

Various mesenchymal cells originating from the lateral plate mesoderm were known to contribute to bone formation, such as growth plate chondrocytes, osteoblasts, and bone marrow stromal cells (BMSCs). In SMN2 1-copy *Smn*^{ΔMPC} mutants, *SMN* may be deleted in all of these cells, suggesting that they all play a role in the bone growth abnormalities observed in 2 weeks mice. However, the population of *Lepr*+*Cxcl12*+ BMSCs, which makes up the majority of bone marrow stromal cells and is positive for *Prrx1*, was reported to be capable of forming new bone only in adults that are over 8 weeks old (Zhou et al., 2014; Matsushita et al., 2020). Consequently, it appears improbable that the observed phenotype at 2 weeks was due to *SMN* deletion in BMSCs. Furthermore, previous researchers revealed that primary osteoblasts from a severe SMA mouse model did not display notable differences from controls in an in vitro ossification test. And they did not observe any differences in bone voxel density and bone thickness in femurs at P3 severe SMA mice. This is supported by the absence of any bone thickness or BMD defects in the SMN2 1-copy mutant (**Figure 1E** and **Figure 1—figure supplement 2A–B**), and the unimpaired osteoblast population (**Figure 1—figure supplement 2E**). Thus, we conclude that the bone growth abnormalities observed in the 2-week-old SMN2 1-copy *Smn*^{ΔMPC} mutant are due to impaired endochondral ossification of growth plate chondrocytes.

To determine whether bone growth defects in SMN2 1-copy *Smn*^{ΔMPC} mutants arise from disrupted chondrocyte homeostasis at growth plates, we stained the femur distal growth plate of P14 mice with hematoxylin and eosin (**Figure 1F**). In line with earlier findings in whole-body SMA mice (Hensel et al., 2020), SMN2 1-copy *Smn*^{ΔMPC} mice exhibited shorter proliferative and hypertrophic zones compared to control mice (**Figure 1G** and **H**). Additionally, there was a significant reduction in the number of chondrocytes in the hypertrophic zone (**Figure 1I**). To investigate the decrease in chondrocyte proliferation and subsequent reduction in the proliferative and hypertrophic zone, we stained the proliferation marker Ki67 in the growth plate of both SMN2 1-copy control and mutant samples (**Figure 1J**). We then quantified the percentage of Ki67+ nuclei in the resting and proliferative zones (**Figure 1K** and **L**). Although there was no significant difference observed in the Ki67+ percentage of proliferative zone chondrocytes in the proceeding proliferation state, there was an absolute reduction in resting zone chondrocyte proliferation. The decreased proliferation rate in the resting zone could have impeded the transition to the proliferative zone. Our data indicate that adequate expression of *SMN* is essential for the homeostasis of chondrocytes at growth plates.

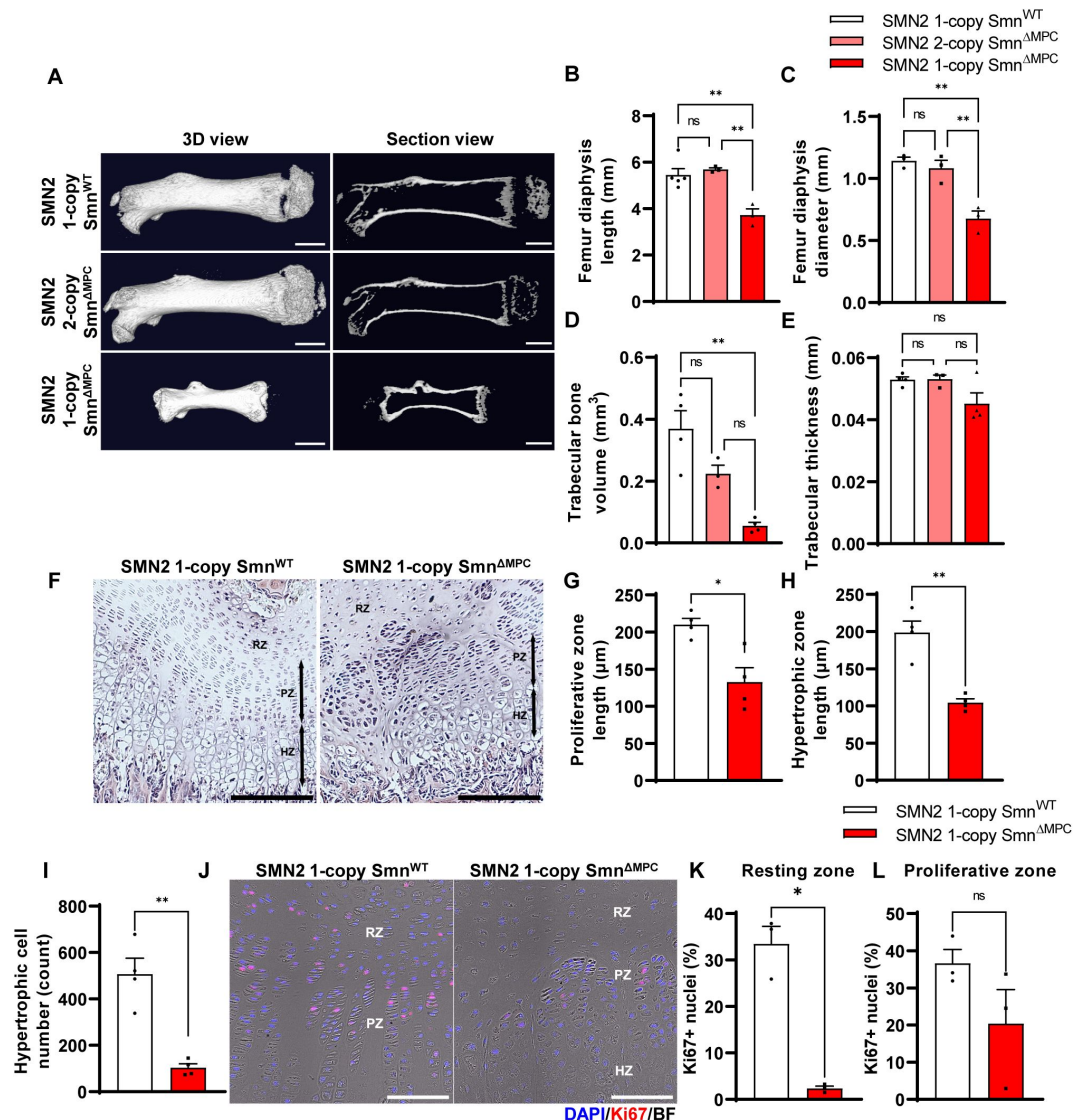


Figure 1

Skeletal growth abnormalities and altered growth plate homeostasis in SMN2 1-copy *Smn*^{ΔMPC} mice.

(A) Representative 3D images and longitudinal section view of the ossified femur bone. Scale bars, 1 mm. (B-C) SMN2 1-copy mutant's femurs showed reduced growth in diaphysis length and diameter, and (D) decreased trabecular bone volume. (E) Trabecular bone thicknesses were not significantly different between the control and mutant groups. The micro CT analysis was performed in femur diaphysis and metaphysis from SMN2 1-copy *Smn*^{ΔWT}, SMN2 2-copy, and 1-copy *Smn*^{ΔMPC} mice at P14. 1-way ANOVA with Tukey's post hoc test, n = 3-5 mice in each genotype (B-E). (F) Representative images of H&E staining in the distal femur growth plate of control and mutant mice with 1 copy of *SMN2* at P14. Scale bars, 100 μm. Resting zone (RZ), Hypertrophic zone (HZ) and proliferative zone (PZ). (G-I) Indicated by black arrows, the HZ and PZ lengths were reduced in SMN2 1-copy *Smn*^{ΔMPC} mice, and the hypertrophic cell number in a section of the 1-copy mutant was decreased (n = 4 mice in each genotype; Unpaired t-test with Welch's correction). (J) Representative images of Ki67 immunostaining in the distal femur growth plate of control and mutant mice with 1 copy of *SMN2* at P14. Scale bars, 100 μm. (K) and decreased Ki67+ percentage in resting zone chondrocytes. (L) Ki67+ percentage in the proliferative zone was not significantly different between the control and mutant groups. n = 3 mice in each genotype; Unpaired t-test with Welch's correction (K-L). ns; not significantly different. *p < 0.05; **p < 0.01. Error bars show s.e.m.

Disruption of chondrocyte-derived IGF

signaling in SMN2 1-copy *Smn*^{ΔMPC} mutants

The proliferation and differentiation of growth plate chondrocytes are regulated by systemic IGF (Shim et al., 2015; Karimian et al., 2012; Shantanam et al., 2018; Racine et al., 2020). Previous research suggested that a key factor contributing to the pathological phenotype in SMA is the lowered expression of the *Igf1/Igfbp3/Igfals* genes, which produce IGF and IGF-carrying proteins, in the liver. (Hua et al., 2011; Murdocca et al., 2012). As the IGF pathway proteins are downregulated in whole-body SMA mice, the bone growth defects observed in the mice have sparked debate as it remains unclear whether they are due to cell-autonomous defects by chondrocyte SMN reduction, or the low liver-derived IGF level (Hua et al., 2011; Tsai et al., 2014; Deguise et al., 2021; Hensel et al., 2015; Hensel et al., 2020).

To clarify this issue, we used *Smn*^{ΔMPC} mutant mice, which enabled us to investigate the effect of SMN depletion in chondrocytes on bone growth, while circumventing the impact of the endocrine signal by *Prrx1*-negative organ. To investigate the impact of IGF signaling on growth plate chondrocytes, we employed immunofluorescence to evaluate the percentage of p-AKT positive cells activated by the IGF-PI3K-AKT pathway in both SMN2 1-copy control and mutant femur distal growth plate (**Figure 2A**). Intriguingly, the percentage of p-AKT⁺ cells was significantly decreased in resting zone chondrocytes, but not in the proliferative zone, which aligns with the Ki67⁺ percentage (**Figure 2B-C**). Expectedly, the liver's mRNA expression of IGF pathway genes, reported to be decreased in SMA mouse models and patients (Hua et al., 2011; Murdocca et al., 2012; Deguise, 2021; Sahashi et al., 2013), showed no difference when comparing controls to SMN2 1-copy *Smn*^{ΔMPC} mutants (**Figure 2D**). These findings indicate that the growth plate's proliferation and hypertrophy in SMN2 1-copy mutants are affected by impairments in another AKT upstream signal rather than by liver-secreted systemic IGF.

There are reports indicating that local IGF expression plays a crucial role in bone development through the growth plate, in addition to circulating IGF (Hallett et al., 2019; Racine et al., 2020). While most *Igf1*-null mice died before birth and had smaller tibial lengths than normal, liver-specific *Igf1*-deleted mice did not experience significant changes in body length or tibial length during postnatal growth, despite a 75% reduction in serum IGF-1 levels (Baker et al., 1993; Yakar et al., 1999). This indicates that local IGF in the growth plate is crucial for endochondral ossification, in addition to serum IGF. The chondrocyte-specific *Igf1* knockout mouse demonstrated a reduction in postnatal body and femur length, and the chondrocyte-specific *Igf1r* knockout mouse demonstrated a significant reduction in bone growth, as well as a decrease in both growth plate proliferative and hypertrophic zone (Govoni et al., 2007; Wang et al., 2011). A recent study revealed that resting zone chondrocytes in the growth plate serve as a major source of local IGF and activate the p-AKT pathway via autocrine and paracrine IGF signaling (Oichi et al., 2023). The study further revealed that cells that constitute bone and bone marrow, apart from chondrocytes, do not express *Igf1*, making chondrocytes the solitary source of local IGF. These suggest that the growth plate defects and the reduction of resting zone AKT phosphorylation in SMN2 1-copy mutants may be due to chondrocyte-secreted IGF deficiency. To confirm this hypothesis, we evaluated the expression of IGF-related genes in chondrocytes from both SMN2 1-copy control and mutant femur (**Figure 2E**). Indeed, *Igf1* and *Igfbp3* were greatly depleted in SMN2 1-copy mutant chondrocytes. It is hypothesized that the increased presence of *Ghr* may be in response to the reduction of *Igf1*. As IGF directly causes chondrocyte hypertrophy (Wang et al., 1999), we assess the mRNA expression of chondrocyte hypertrophic marker *Col10a1* and undifferentiated chondrocyte marker *Col2a1* (**Figure 2F**). The results show that *Col10a1* is decreased, while *Col2a1* is increased in the mutant. In **Figure 1H-I**, the hypertrophic cell reduction may be caused by a low local IGF level. Therefore, depletion of local IGF in the growth plate of SMN2 1-copy mutants may hinder chondrocyte progression to proliferation and hypertrophy, leading to aberrations in bone development. In **Figures 1L** and **2C**, it is possible that the serum IGF could affect the remnant proliferative and p-AKT-positive cells in the growth

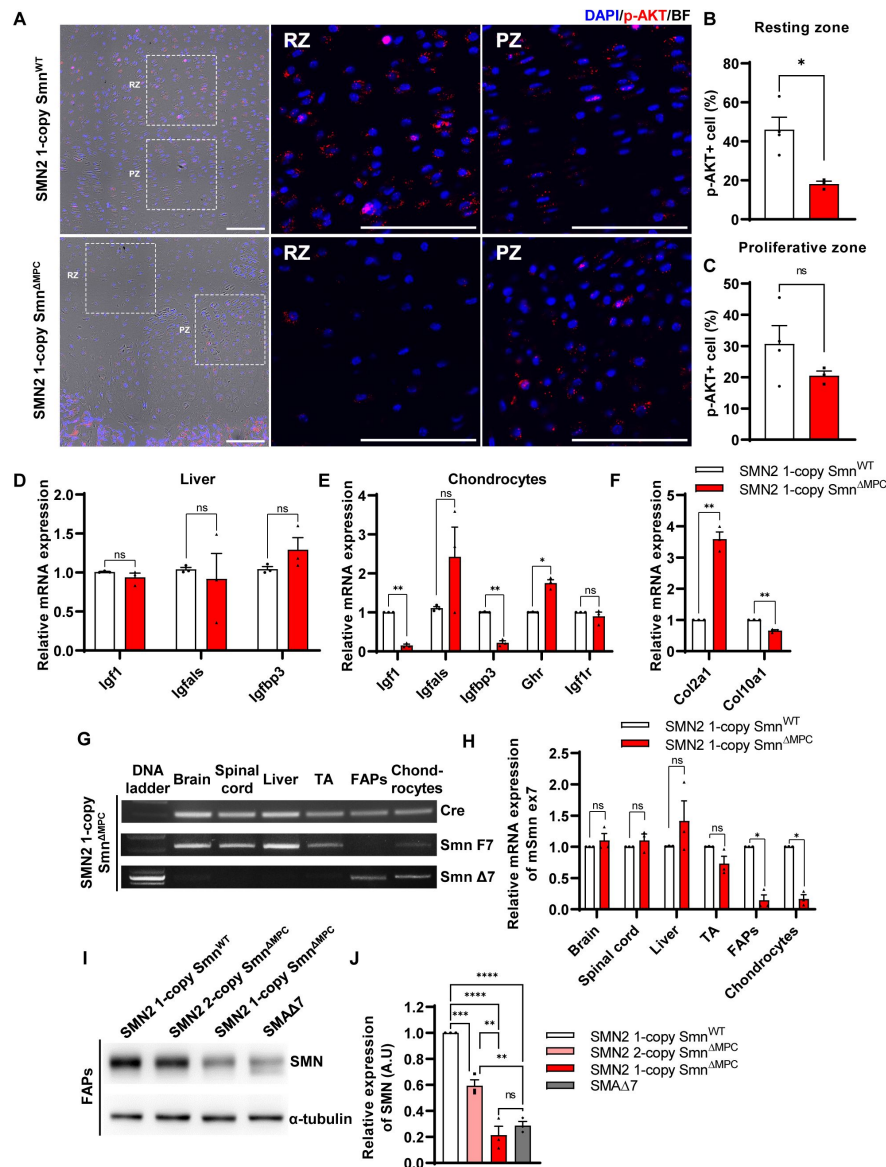


Figure 2

Decreased chondrocytes-derived IGF-AKT axis by limb mesenchymal cell-specific SMN depletion in SMN2 1-copy *Smn*^{ΔMPC} mice.

(A) Representative images of p-AKT immunostaining in distal femur growth plate from mice at P14. Scale bars, 100 μ m. (B-C) The p-AKT-positive percentage was decreased in the resting zone chondrocytes, not in the proliferative zone. ($n = 3-4$ mice in each genotype; Unpaired t-test with Welch's correction) (D) Relative IGF axis mRNA expression in the livers of SMN2 1-copy control and mutant mice. The IGF pathway genes showed no difference when comparing controls to SMN2 1-copy *Smn*^{ΔMPC} mutants. (E-F) Relative IGF axis and chondrocyte differentiation marker mRNA expression in the chondrocytes of SMN2 1-copy control and mutant mice. The *Igf1*, *Igfbp3*, and hypertrophic marker *Col10a1* expression were decreased in SMN2 1-copy *Smn*^{ΔMPC} mutants. $n = 3$ mice in each genotype; Unpaired t-test with Welch's correction (D-F). (G) Representative images of genomic PCR analysis from SMN2 1-copy *Smn*^{ΔMPC} mice tissues at P21. (H) qRT-PCR analysis from tissues of SMN2 1-copy *Smn*^{WT} and *Smn*^{ΔMPC} mice at P21 ($n = 3$ mice in each genotype; Unpaired t-test with Welch's correction). Deletion of *Smn* exon 7 was detected only in limb mesenchymal cells using genomic PCR (G) and full-length SMN mRNA expression (H). (I) Representative images of western blot analysis in cultured FAPs. SMN protein in FAPs of SMN2 1-copy *Smn*^{ΔMPC} mice exhibited a decrease comparable to that observed in the SMAΔ7 mice. (J) Relative SMN levels in cultured FAPs of the controls and mutants ($n = 3$ mice in each genotype; 1-way ANOVA with Tukey's post hoc test). ns; not significantly different. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$. Error bars show s.e.m.

plate via the vascularization of bone marrow. Based on these findings, we concluded that the bone growth defects observed in the whole-body SMA mouse model are mainly due to a depletion of localized IGF secretion by SMN-deleted chondrocytes.

Mesenchymal cell-specific SMN reduction similar to severe

SMA mouse model in SMN2 1-copy *Smn*^{ΔMPC} mutants

To confirm the specific deletion of *Smn* in limb mesenchymal cells, including chondrocytes and FAPs, we performed qRT-PCR for full-length SMN mRNA expressed by undeleted *Smn* and the SMN2 alternative splicing in each tissue. Our findings indicate that full-length SMN mRNA expression in the brain, liver, skeletal muscle, and spinal cord of SMN2 1-copy *Smn*^{ΔMPC} mice at postnatal day 21 (P21) was similar to that of control mice, while it was significantly reduced in isolated FAPs and chondrocytes (**Figure 2H** [↗](#)). Additionally, we confirmed the presence of the *Smn*Δ7 variant, which is the exon 7-deleted form of *Smn* created by cre-lox mediated recombination, in both FAPs and chondrocytes by conducting genomic PCR on the SMN2 1-copy *Smn*^{ΔMPC} mutant (**Figure 2G** [↗](#)). Since SMN was not down-regulated in the tissues other than limb mesenchymal cells, non-mesenchymal cells were ruled out from being responsible for the phenotype observed in *Smn*^{ΔMPC} mutants.

Among our *Smn*^{ΔMPC} models, mice with two copies of SMN2 exhibit similar bone development parameters as control mice without SMN deletion. However, mice with one copy of SMN2 display bone pathological defects akin to SMA mouse models. This may be because the quantity of human full-length SMN protein produced by SMN2 2-copy, was sufficient to sustain SMN complex function in the SMN2 2-copy *Smn*^{ΔMPC} mutants, despite the absence of functional mouse SMN protein in limb mesenchymal cells. On the contrary, due to the lower expression of full-length SMN compared to SMN2 2-copy mutants, the SMN complex may not function properly in SMN2 1-copy mutant cells. We confirmed this by comparing the amount of full-length SMN protein in isolated FAPs from the hindlimbs of control, *Smn*^{ΔMPC} mutants, and a severe SMA mouse model (*Smn*^{-/-}; *SMN2*^{+/+}; *SMN*Δ7^{+/+}; -SMAΔ7 mutants -) (**Figure 2I** [↗](#)). The data show that SMN2 1-copy *Smn*^{ΔMPC} mutants exhibited ~80% reduction in SMN levels compared to the control group. The level of SMN protein in SMN2 1-copy *Smn*^{ΔMPC} mutants was similar to that in SMAΔ7 mutants. Conversely, SMN2 2-copy mutants display a decrease of approximately 40% in SMN protein levels compared to the control (**Figure 2J** [↗](#)). The moderately reduced expression of SMN is adequate to support regular bone development in SMN2 2-copy mutants. Taken together, our findings indicate that the reduction of mesenchymal SMN to levels comparable to that of the severe SMA mouse model causes SMA-like bone pathology in the SMN2 1-copy mutant.

Abnormal NMJ maturation in SMN2 1-copy *Smn*^{ΔMPC} mutants

To investigate whether disabling SMN in FAPs results in SMA-like neuromuscular impairments, we assessed if NMJ phenotypes observed in SMA mouse models also occur in SMN2 1-copy *Smn*^{ΔMPC} mutants. Both severe and mild SMA mouse models exhibit impaired NMJ maturation markers, including plaque-like morphology of acetylcholine receptor (AChR) clusters, neurofilament (NF) varicosities, and poor terminal arborization (Kong et al., 2009 [↗](#); Martinez, 2012; Monani et al., 2003 [↗](#); Kariya et al., 2008 [↗](#)). To evaluate the impact of SMN deficiency in FAPs on NMJ maturation, we evaluated the NMJ maturation markers in the TA muscles of control and SMN2 1-copy *Smn*^{ΔMPC} mutant mice at P21, a time when NMJ maturation is in progress (**Figure 3A** [↗](#)). Our examination revealed the presence of NF varicosities in SMN2 1-copy mutants as compared with control mice (**Figure 3A** [↗](#) and **3D** [↗](#)). Additionally, the number of nerve branches was decreased and half of the total NMJs were poorly arborized in SMN2 1-copy mutants (**Figure 3B** [↗](#)-**C** [↗](#)). These presynaptic alterations are specific phenotypes in neurogenic atrophy like SMA. Unlike neurogenic atrophy, physiologic atrophy shows no differences in presynaptic morphology, such as nerve branching (Deschenes et al., 2006 [↗](#)). This suggests that the NMJ phenotypes observed in SMN2 1-copy *Smn*^{ΔMPC} mutant mice are not caused by decreased muscle size and activity resulting

from bone growth abnormalities. The morphology of AChR clusters shows that the mutants have more immature plaque-like NMJs than the controls' pretzel-like structure (**Figure 3E** [↗](#)). Therefore, these findings indicate that SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutants exhibit NMJ maturation abnormalities common in SMA mouse models.

Undisturbed NMJ formation in neonatal

SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutants

To determine whether any NMJ defects were present prior to juvenile NMJ maturation in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$, we examined NMJ formation in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice at the neonatal stage on postnatal day 3. We evaluated the AChR and nerve terminal areas to assess postsynaptic and presynaptic development, respectively (**Figure 3F** [↗](#)). Measurements of AChR cluster size indicated no differences between control and SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice (**Figure 3G** [↗](#)). However, the area of AChR covered by nerve terminals was slightly larger in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ (**Figure 3H** [↗](#)). We have no reasonable explanation for why the coverage is higher in the mutant. However, there does not appear to be abnormal development of the NMJ in the mutant, at least until the neonatal period. Therefore, we reasoned that SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutants began to exhibit deterioration in the NMJ maturation during the juvenile stage, following the intact neonatal development of NMJ.

Aberrant NMJ morphology in the adult SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice

To evaluate the organization of NMJ after the conclusion of postnatal NMJ development, considering mesenchymal SMN expression, we examined NMJ morphology in the TA muscle of control, SMN2 1-copy, and 2-copy $\text{Smn}^{\Delta\text{MPC}}$ mice at postnatal day 56 (P56). Our analysis revealed that presynapses were fragmented in SMN2 1-copy mutants, resulting in a bouton-like morphology, in contrast to the control and SMN2 2-copy $\text{Smn}^{\Delta\text{MPC}}$ mice (**Figure 4A** [↗](#)). In SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice, a two-fold presynaptic fragmentation compared to the control was quantified, demonstrating nerve terminal shrinkage (**Figure 4B** [↗](#)). Additionally, NF ends displayed more severe varicosities than at P21 and were only connected to the proximal nerve by very thin NF, unlike control and SMN2 2-copy mice (**Figure 4C** [↗](#)). Remarkably, numerous presynaptic islands formed in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice through the merging of fragmented presynapses and NF varicosity. In SMN2 1-copy mutants, AChR clusters displayed fragmented grape-shaped morphology that overlapped with nerve terminals, whereas control and SMN2 2-copy mice displayed pretzel-like structures (**Figure 4D** [↗](#)). These results suggest that defects in adult NMJ morphology occur when mesenchymal SMN protein is reduced to the extent of the SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutants.

Presynaptic neurotransmission alteration

in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutants

To investigate whether the morphologically aberrant NMJs of SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice have functional impairments, we isolated hindlimb extensor digitorum longus (EDL) muscles from P56 mice and conducted electrophysiological recording. We incubated the muscles with μ -conotoxin, which selectively inhibits muscle voltage-gated Na^+ channels, prevented the induction of muscle action potential (Ling et al., 2010 [↗](#); Zanetti 2018 [↗](#)), and recorded the Miniature endplate potential (mEPP) and evoked endplate potential (eEPP). (Ling et al., 2010 [↗](#); Zanetti 2018 [↗](#)). mEPP, a response that occurs when spontaneously released acetylcholine binds to nicotinic AChR without nerve stimulation, was measured ex vivo near the NMJs of the EDL muscles in the control and SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice (**Figure 5A** [↗](#)). The mEPP amplitude was increased in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice (**Figure 5B** [↗](#)), whereas mEPP frequency was comparable between the controls and mutants (**Figure 5C** [↗](#)). The results indicate that the NMJ synapses of SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice

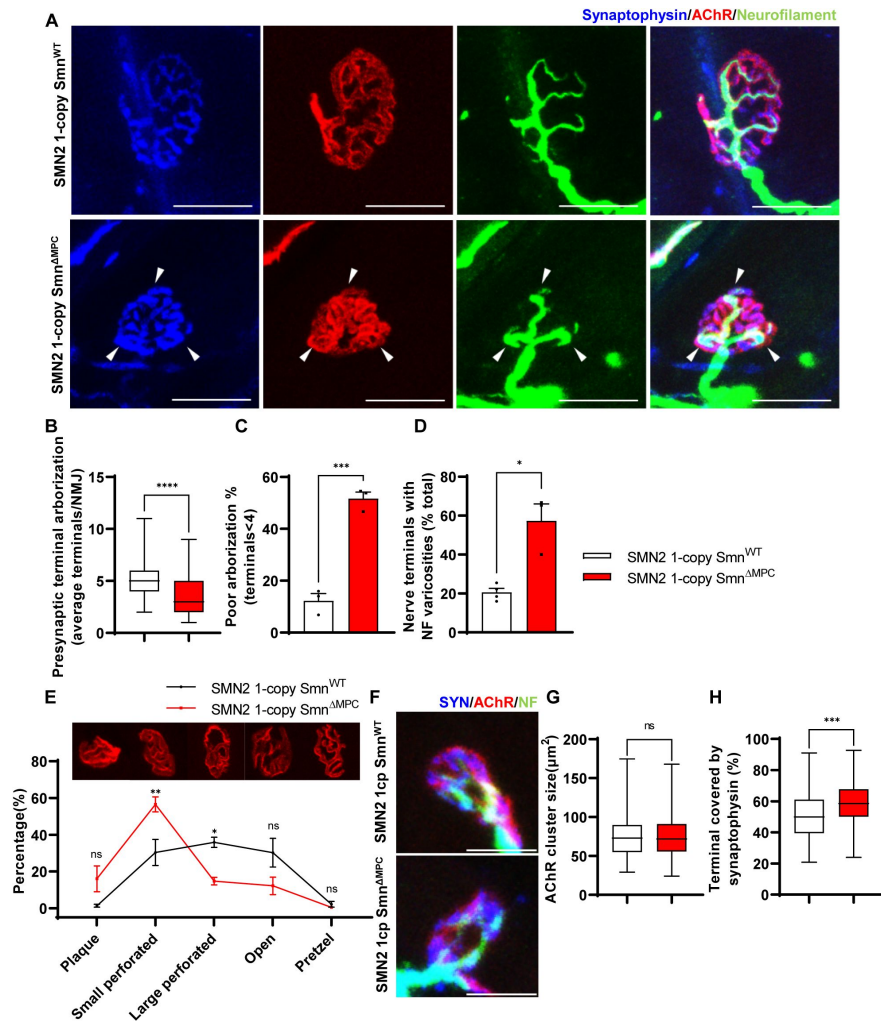


Figure 3

Aberrant postnatal NMJ maturation in SMN2 1-copy Smn^{ΔMPC} mice.

(A) Immunostaining of NMJs in TA muscle of SMN2 1-copy Smn^{WT} and Smn^{ΔMPC} mice at P21 with anti-NF (Green), anti-synaptophysin (Blue), α-Btx staining AChR (Red). Scale bars, 20 μm. The confocal images of NMJs showed decreased presynaptic terminal branching and the existence of nerve terminal varicosities that were enlarged with NF (indicated by arrowheads) in the mutant. (B) The NMJs of the SMN2 1-copy mutant exhibited a significant decrease in presynaptic terminal arborization and (C) an increased percentage of poorly arborized NMJs ($n = 3$ mice in each genotype; Unpaired t-test with Welch's correction). (D) The percentage of NMJs exhibiting NF varicosities was higher in the SMN2 1-copy mutant group than in the control group ($n = 3$ -4 mice in each genotype; Unpaired t-test with Welch's correction). (E) For quantification of the NMJ maturation stage, we classified NMJs into five distinct developmental stages (Plaque: Plaque-shaped endplate without any perforation; Small perforated: Plaque-shaped endplate with small perforations; Large perforated: Plaque-shaped endplate with large perforations; Open: C-shaped endplate; Pretzel: Pretzel-like shaped endplate) and then compared the frequency patterns of SMN2 1-copy control and mutant mice ($n = 3$ mice in each genotype; 2-way ANOVA with Tukey's post hoc test). The NMJs of SMN2 1-copy mutants displayed plaque-like shapes, indicating that they were in the immature stage. (F) Immunostaining of NMJs in TA muscle of SMN2 1-copy Smn^{WT} and Smn^{ΔMPC} mice at P3 with anti-NF (Green), anti-synaptophysin (Blue), α-Btx staining AChR (Red). Scale bars, 10 μm. (G) There were no significant differences in AChR cluster size between the SMN2 1-copy control and mutant at P3 ($n = 3$ -4 mice in each genotype; Unpaired t-test with Welch's correction). (H) The ratio of the Synaptophysin area to the AChR area in NMJ was slightly higher in the SMN2 1-copy mutant at P3 ($n = 3$ -4 mice in each genotype; Unpaired t-test with Welch's correction). ns; not significantly different. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$. All box-and-whisker plots show the median, interquartile range, minimum, and maximum. For the box-and-whisker plots, range bars show minimum and maximum (B, G, H). For the bar and line graph, error bars show s.e.m (C, D, E).

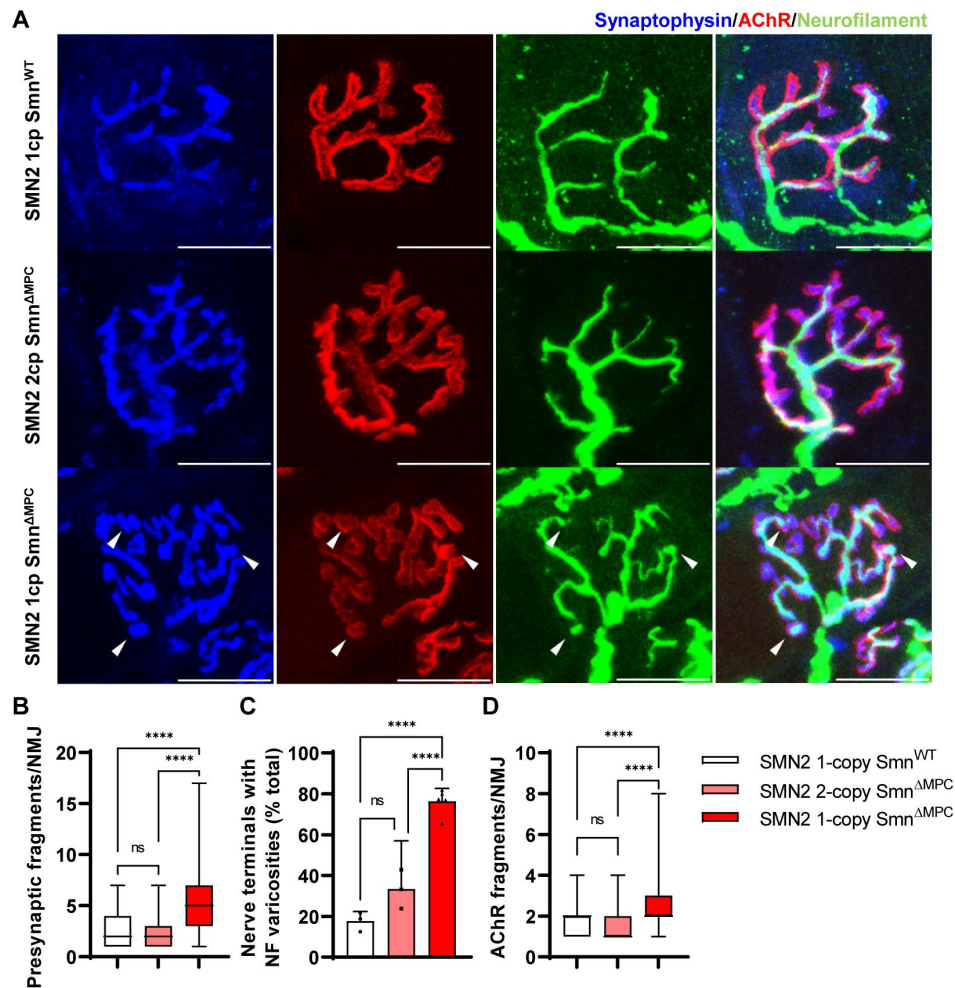


Figure 4

Morphological deterioration in NMJs of adult SMN2 1-copy $Smn^{\Delta MPC}$ mice.

(A) Immunostaining of NMJs in TA muscle of SMN2 1-copy Smn^{WT} , SMN2 2-copy, and SMN2 1-copy $Smn^{\Delta MPC}$ mice at P56 with anti-NF (Green), anti-synaptophysin (Blue), α -Btx staining AChR (Red). Scale bars, 20 μ m. The confocal images of NMJs showed fragmentation and bouton-like NF varicosities (indicated by arrowheads) in the SMN2 1-copy $Smn^{\Delta MPC}$ mice. (B-D) The NMJs of the SMN2 1-copy mutant displayed fragmented presynapse (B), endplate (D), and NF varicosities (C) compared to SMN2 1-copy Smn^{WT} and SMN2 2-copy $Smn^{\Delta MPC}$ mice ($n = 3-5$ mice in each genotype; Presynaptic fragments & AChR fragments: Brown–Forsythe and Welch ANOVA with Games-Howell’s test; NF varicosities: 1-way ANOVA with Tukey’s post hoc test). ns; not significantly different. **** $p < 0.0001$. All box-and-whisker plots show the median, interquartile range, minimum, and maximum (B, D). For the bar graph, error bars show s.e.m. (C)

are functional and more sensitive to acetylcholine compared to the controls. Next, we measured the eEPP by stimulating an action potential at the peroneal nerve (**Figure 5D**). Despite the increased mEPP amplitude, the amplitude of eEPPs was significantly decreased in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice (**Figure 5E**). These results suggest that the nerve terminals in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice exhibit decreased quantal content. This could be due to a decrease in vesicle release probability. Notably, there was no difference in the paired-pulse response, indicating normal neurotransmitter release probability (**Figure 5F–G**). Taken together, these findings suggest that the presynaptic neurotransmission ability of the NMJ is reduced in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutants.

Disturbed nerve terminal structure in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice

To examine the NMJ ultrastructure of SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutants, we utilized transmission electron microscopy (TEM) (**Figure 6A**). The density of junctional folds in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutant specimens was comparable to that of the control (**Figure 6B**). However, the density of synaptic vesicles was substantially elevated in the SMN2 1-copy mutants (**Figure 6C**). Since previous electrophysiological results suggested a decrease in presynaptic neurotransmission capacity in SMN2 1-copy mutants, this could be due to synaptic vesicles failing to fuse with the membrane, leading to the accumulation of vesicles in the terminal and reduced quantal contents in **Figure 5**. In **Figure 3H**, larger synaptophysin coverage in the mutant may be caused by this synaptic vesicle accumulation. Additionally, the detachment of the nerve terminal is more frequent at the NMJ of mutants (**Figure 6D**). The detachment of nerve terminals observed in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutants could have also resulted in diminished presynaptic neurotransmission capacity. Collectively, these findings indicate that SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutants have nerve terminal-specific pathological defects at the NMJ ultrastructural level.

FAPs transplantation rescues NMJ morphology in limb mesenchymal SMN mutants

SMN-deleted limb mesenchymal tissues in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutants comprise not only FAPs, but also bone, cartilage, pericytes, and tendon, among others (Leinroth et al., 2022; Nassari et al., 2017). To evaluate the critical role of FAPs in the postnatal development of the NMJ, we isolated fluorescent protein-labeled wild-type FAPs from $\text{Prrx1}^{\text{Cre}}; \text{Rosa26}^{\text{YFP/+}}$ or $\text{Prrx1}^{\text{Cre}}; \text{Rosa26}^{\text{tdTomato/+}}$ mice and transplanted them into the TA muscles of SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice at postnatal day 10. The TA muscle on the contralateral side was treated with a vehicle as a control. In SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice at P56, the muscle that received FAPs showed decreased presynaptic fragmentation, NF varicosities, and postsynaptic fragmentation compared to the contralateral muscle (**Figure 7A–D**). As a result, the transplanted muscle was not significantly different from the control except for NF varicosities. These data demonstrate that the transplantation of wild-type FAPs rescues the abnormal NMJ development in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice. Overall, our findings indicate that SMN depletion in FAPs leads to the neuronal SMN-independent NMJ pathology in severe SMA, which is rescued by the transplantation of healthy FAPs.

Discussion

In this paper, we elucidate the contribution of SMN depletion in mesenchymal progenitors for the pathogenesis of SMA. To test this hypothesis, we generated conditional knockout mouse strains to delete the *Smn* allele specifically in limb mesenchymal cells and carry human *SMN2* copies. Our research using these mouse models resulted in three major discoveries. First, SMN deficiency in FAPs contributes to NMJ pathological defects in SMA. We observed delayed NMJ maturation and

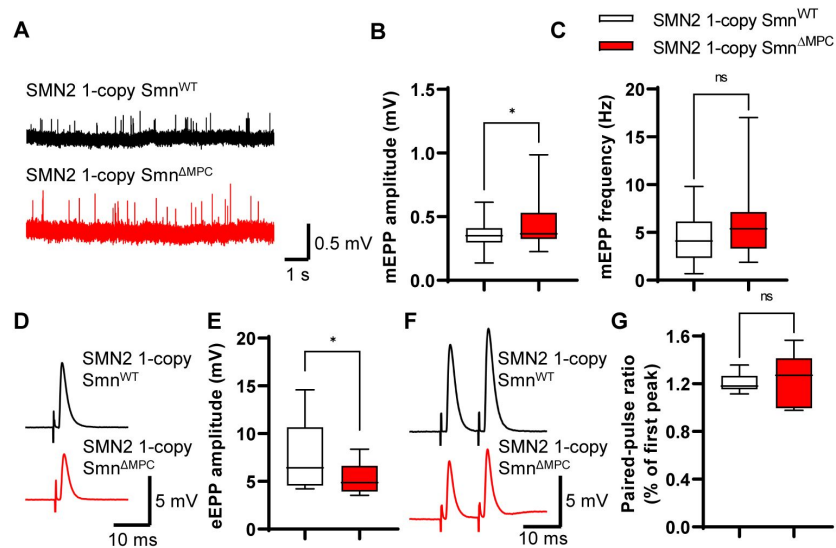


Figure 5

Reduced presynaptic neurotransmission ability in the NMJs of SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mice.

(A) Representative traces of mEPP from SMN2 1-copy $\text{Smn}^{\Delta\text{WT}}$ (top) and SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ (bottom) mice. (B–C) SMN2 1-copy mutant's NMJs showed an increase in mEPP amplitude and no differences in mEPP frequency (1-copy control, $n = 25$, 9 mice; 1-copy mutant, $n = 21$, 8 mice; Unpaired t-test with Welch's correction). (D) Representative traces of eEPP from SMN2 1-copy $\text{Smn}^{\Delta\text{WT}}$ (top) and SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ (bottom) mice. (E) The mutant's NMJs showed a stronger amplitude of eEPPs (1-copy control, $n = 12$, 4 mice; 1-copy mutant, $n = 12$, 3 mice; Unpaired t-test with Welch's correction). (F) Representative traces of paired-pulse response from SMN2 1-copy $\text{Smn}^{\Delta\text{WT}}$ (top) and SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ (bottom) mice. (G) Paired-pulse response was not different between SMN2 1-copy control and mutant NMJs, indicating a comparable neurotransmitter release probability (1-copy control, $n = 8$, 3 mice; 1-copy mutant, $n = 6$, 3 mice; Unpaired t-test with Welch's correction). The electrophysiological recording was performed in the EDL muscle at P56. ns: not significantly different. $*p < 0.05$. All box-and-whisker plots show the median, interquartile range, minimum, and maximum. For the box-and-whisker plots, range bars show minimum and maximum (B, C, E, G).

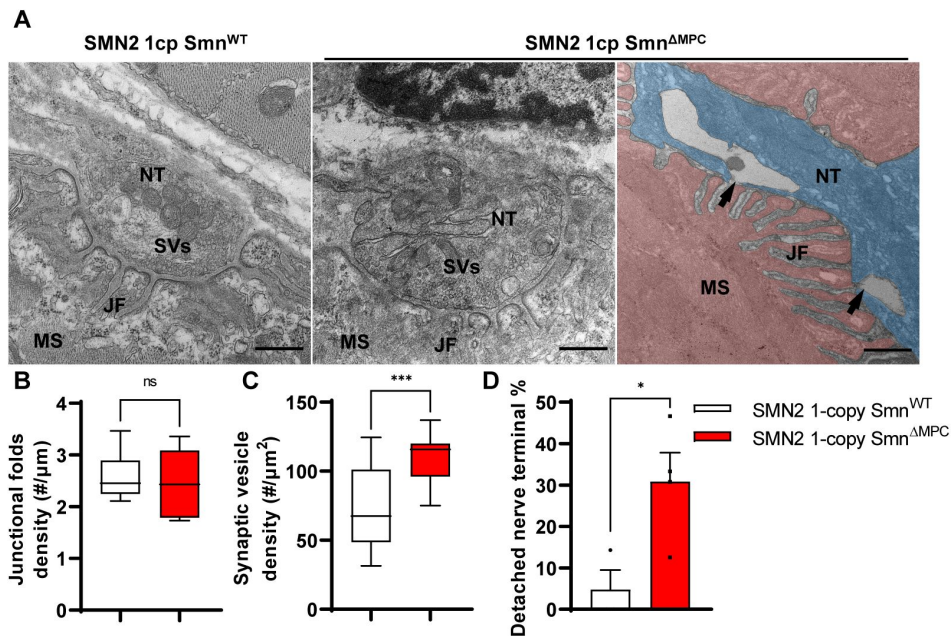


Figure 6

Abnormal nerve terminal ultrastructure in SMN2 1-copy mutant.

(A) Representative TEM images from NMJs of SMN2 1-copy Smn^{ΔWT} and SMN2 1-copy Smn^{ΔMPC} mice at P56. Scale bars, 500 nm. Nerve terminal (NT; indicated by the blue zone). Synaptic vesicles (SVs). Muscle fiber (MS; indicated by the red zone). Endplate junctional folds (JF). Nerve terminal detachment (indicated by arrow) was observed in SMN2 1-copy Smn^{ΔMPC} mice. **(B)** The density of junctional folds in the NMJ of SMN2 1-copy Smn^{ΔMPC} mice no significant change compared to the control, whereas **(C)** the density of synaptic vesicles was increased ($n = 3-4$ mice in each genotype; Unpaired t-test with Welch's correction). **(D)** The detachment of the nerve terminal occurs more frequently at the NMJ of mutants ($n = 3-4$ mice in each genotype; Unpaired t-test with Welch's correction). ns; not significantly different. $*p < 0.05$; $***p < 0.001$. All box-and-whisker plots show the median, interquartile range, minimum, and maximum. For the box-and-whisker plots, range bars show minimum and maximum (**B, C**). For the bar graph, error bars show s.e.m(**D**).

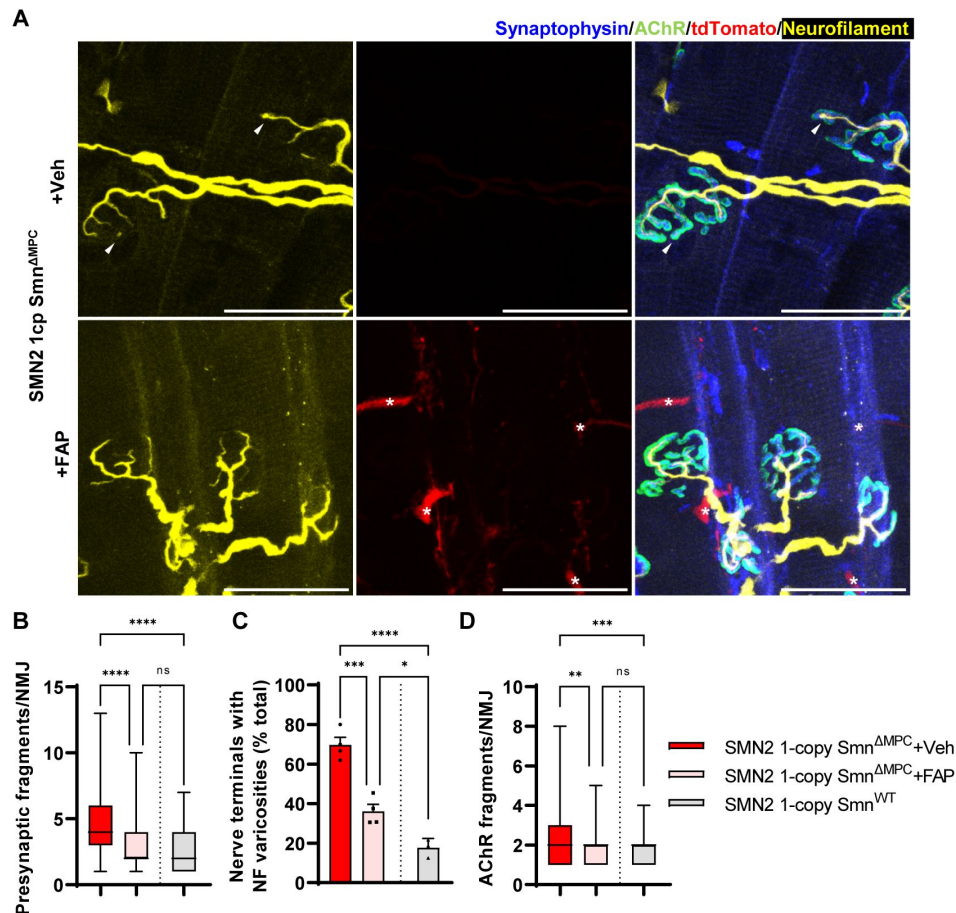


Figure 7

Improved postnatal NMJ development in the TA muscle of SMN2 1-copy Smn^{ΔMPC} mice following healthy FAPs transplantation.

(A) Immunostaining of NMJs in tdTomato⁺ FAPs-transplanted TA muscle (+FAP) and vehicle-treated contralateral muscle (+Veh) in SMN2 1-copy Smn^{ΔMPC} mice at P56 with anti-NF (Yellow), anti-synaptophysin (Blue), α-Btx staining AChR (Green), and tdTomato fluorescence (Red). Scale bars, 40 μm. The images revealed NF varicosities (indicated by arrowheads) in the +Veh NMJs. The tdTomato⁺ FAPs (marked by asterisks) were transplanted into +FAP NMJs, which exhibited (B) decreased presynaptic fragmentation, (C) NF varicosities, and (D) AChR fragmentation compared to +Veh NMJs and similar to wild-type NMJs. (n = 3-4 mice in each group; Unpaired t-test with Welch's correction). ns; not significantly different. ***p* < 0.01, ****p* < 0.001, *****p* < 0.0001. All box-and-whisker plots show the median, interquartile range, minimum, and maximum. For the box-and-whisker plots, range bars show minimum and maximum (B, D). For the bar graph, error bars show s.e.m (C).

varicosities in juvenile SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutant. The pathogenic NMJ phenotypes were also observed in the SMA $\Delta 7$ mutant, which is one of the severe SMA mouse models (Kong et al., 2009; Martinez, 2012; Kariya et al., 2008). As the SMA $\Delta 7$ mutant typically lives for approximately 12 days, the fragmentation of the NMJ in adult SMN2 1-copy mutant was not evaluated in the severe SMA mutant. Nevertheless, models that induce a conditional adult SMN deficiency through either Cre^{ER} allele or oligonucleotide administration resulting in SMN reduction, demonstrated the depletion of SMN throughout the body caused fragmentation of NMJ (Sahashi et al., 2013; Kariya et al., 2014). Thus, we demonstrate that the SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutant model mimics whole-body SMA mouse models in NMJ morphology. In the electrophysiological test, the SMA $\Delta 7$ mutant exhibited reduced quantal content, readily releasable pool, and vesicle release probability (Torres-Benito et al., 2011). In our study, we did not directly assess the readily releasable pool in SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutant by stimulus train of electrophysiological recording but instead showed reduced quantal content and normal vesicle release probability. In previous studies, it was theorized that the decrease in quantal contents in SMA $\Delta 7$ mutants resulted from decreased synaptic vesicle density at nerve terminals caused by motor neuron defects and abnormal axonal transport (Dale et al., 2011; Kong et al., 2009). However, we found that synaptic vesicle density was increased in the SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutant with SMN-sufficient motor neurons. It is possible that the alteration of active zones, which were also altered in the SMA motor terminals (Kong et al., 2009), contributed to the reduction in synaptic vesicle fusion and the decreased quantal contents. Indeed, nerve terminal detachment in SMN2 1-copy mutant mice was also found in active zone complex protein integrin- $\alpha 3$ knockout mice (Ross et al., 2017). Based on the rescue data of transplanted healthy FAPs, we can report that FAPs-specific SMN depletion is involved in NMJ pathology of SMA.

Second, we demonstrated that skeletal growth defects, a phenotype observed in SMA (Khatri et al., 2008; Vai et al., 2015; Wasserman et al., 2017; Baranello et al., 2019; Hensel et al., 2020), are a cell-autonomous pathological effect of the depletion of SMN in chondrocytes. We observed reduced bone size and volume in juvenile SMN2 1-copy $\text{Smn}^{\Delta\text{MPC}}$ mutant. Depletion of SMN from chondrocytes is sufficient to cause growth problems with chondrocyte-secreted IGF defects, which affects growth plate homeostasis. Previous studies have reported that *IGF1* overexpression improves biochemical and behavioral manifestations in SMA mice, suggesting potential therapies for SMA (Bosch-Marce et al., 2011; Tsai et al., 2014). Our study showed that these IGF therapies for SMA could be one to consider for treating bone growth abnormalities. In addition, the skeletal growth defect may affect physiologic muscle atrophy through imbalanced muscle contraction and reduced tension. This physiologic atrophy may be part of SMA-associated muscle weakness independent of neurogenic atrophy.

Third, it was demonstrated that adequate levels of SMN protein are essential for MPCs to contribute to limb neuromusculoskeletal development. The mutants with only 1 copy of *SMN2* exhibit problematic symptoms observed in both SMA patients and mouse models, while SMN2 2-copy mutants display a typical phenotype in bone and NMJ. The lack of SMN protein in MPCs by insufficient *SMN2* copies, similar to the deficiency seen in severe SMA, is responsible for the onset of SMA pathology. Thus, we propose that restoration of deficient SMN in MPCs is crucial for rehabilitating their function. Based on these discoveries, SMN replenishment treatments for MPCs, specifically FAPs, and chondrocytes, are necessary to provide a complete solution for neuromusculoskeletal defects in severe SMA patients.

The initial focus for the treatment of SMA was on motor neurons located in the spinal cord, and pharmaceuticals were created to address the deficiency of SMN in these neurons (Mercuri et al., 2020). For example, intrathecal injections of Spinraza can efficiently boost SMN in the CNS, including the spinal cord (Passini et al., 2011; Claborn et al., 2018). However, Spinraza does not address the lack of SMN in peripheral tissues, including mesenchymal cells, highlighted in recent studies. The drug Zolgensma, which employs AAV9 to express SMN through systemic delivery, appears to resolve this issue (Foust et al., 2009; Foust et al., 2022; Valori et al., 2010; Mattar

et al., 2013 [\[13\]](#)). Nevertheless, a prior study indicates that chondrocytes within the growth plate and articular cartilage do not get infected with AAV9 (Yang et al., 2019 [\[14\]](#)). Furthermore, it is well-established that DNA vectors delivered via AAV9 undergo dilution during cell proliferation (Penaud-Budloo et al., 2008 [\[15\]](#); Colella et al., 2018 [\[16\]](#); Van-Alstyne et al., 2021; Heller et al., 2021 [\[17\]](#)). Given the infection issue and the dilution issue by the active cell population changes of chondrocytes and FAPs during postnatal development (Petrany., 2020 [\[18\]](#); Bachman et al., 2022 [\[19\]](#)), SMN supplementation through Zolgensma alone would not be sufficient for severe SMA patients. Thus, there is an urgent need to research and develop therapeutic strategies that target mesenchymal progenitors.

While this study is the first to demonstrate the impact of SMN depletion in FAPs on the NMJ development, it does not elucidate the specific mechanism by which FAPs influence the NMJ. Our study observed NMJ recovery in the muscles transplanted with healthy FAPs, but not in the contralateral muscles, indicating that FAPs are likely involved in NMJ development through juxtacrine or paracrine signaling. Since FAPs interact with surrounding tissues through a variety of signaling factors, such as extracellular matrix (Contreras et al., 2021 [\[20\]](#); Scott et al 2019 [\[21\]](#)), Wnt-related protein (Lukjanenko et al., 2019), and Bmp signaling protein (Uezumi et al., 2021 [\[22\]](#); Camps et al., 2020 [\[23\]](#)), mis-splicing of the signaling factors due to SMN reduction could disturb the homeostasis of neighboring tissues (Zhang et al., 2008). Furthermore, the *Hsd11b1*-positive subpopulation of FAPs associated with NMJ was discovered through single-cell RNA sequencing in a recent study (Leinroth et al., 2022 [\[24\]](#)). This population is located adjacent to the NMJ and responds to denervation, indicating an increased possibility of interaction with the NMJ organization. Therefore, it is necessary to conduct additional investigations into the expression of various signaling factors by diverse FAP subpopulations in future studies.

Methods

Animals

Prrx1^{Cre} (stock 005584), *Smn^{f7/+}* (stock 006138), *Rosa26^{YFP/+}* (stock 006148), *Rosa26^{tdTomato/+}* (stock 007914), *SMN2^{+/+}*, *Smn^{+/-}* (stock 005024 – *Smn* knockout and *SMN2* homologous transgenic mouse –) and *Smn^{+/-}; SMN2^{+/+}; SMNΔ7^{+/+}* (stock 005025 005024 – *Smn* knockout and *SMN2*, *SMNΔ7* homologous transgenic mouse –) mice were acquired from The Jackson Laboratory (Bar Harbor, ME, USA). *SMN2* 0-copy *Smn^{ΔMPC}* mice (*Prrx1^{Cre}; Smn^{f7/f7}*), *SMN2* copy *Smn^{ΔMPC}* mice (*Prrx1^{Cre}; Smn^{f7/f7}; SMN2^{+/-}* – *SMN2* heterologous allele–), and *SMN2* copy *Smn^{ΔMPC}* mice (*Prrx1^{Cre}; Smn^{f7/f7}; SMN2^{+/-}* – *SMN2* homologous allele–), were generated by crossing *Prrx1^{Cre}* mice with *Smn^{f7/+}* and *Smn^{+/-}; SMN2^{+/+}* mice. To utilize fibro-adipogenic progenitors (FAPs) transplantation, we generated *Prrx1^{Cre}; Rosa26^{YFP/+}* mice and *Prrx1^{Cre}; Rosa26^{tdTomato/+}* mice by breeding *Prrx1^{Cre}* with *Rosa26^{YFP/+}* and *Rosa26^{tdTomato/+}* mice, respectively. To avoid deletion of the floxed allele by *Prrx1^{Cre}* expression in female germline, we bred females without *Prrx1^{Cre}* line to *Prrx1^{Cre}* transgenic males. *SMAΔ7* mutants (*Smn^{-/-}; SMN2^{+/-}; SMNΔ7^{+/-}*) were produced by mating *Smn^{+/-}; SMN2^{+/-}; SMNΔ7^{+/-}* mice. Both male and female mice were used in the experiments, and no sex-specific differences were observed. Control littermates lacking *Prrx1^{Cre}* were utilized for analysis. All mouse lines were housed under controlled conditions with specific pathogen-free and handled according to the guidelines of the Seoul National University Institutional Animal Care and Use Committee (IACUC).

Micro CT

The femurs from three groups - control, *SMN2* 2-copy *Smn^{ΔMPC}*, and *SMN2* 1-copy *Smn^{ΔMPC}* mice at postnatal day 14 (P14) - were isolated and cleaned of muscles and skin. Subsequently, the femurs were preserved in 4% paraformaldehyde (PFA) in PBS overnight at 4°C before micro-computed tomography (micro CT). The femurs were then imaged through a Quantum GX II micro-CT imaging system (PerkinElmer, Hopkinton, MA, USA). The X-ray source for scanning was set at

90 kV and 88 mA with a field of view of 10 mm (voxel size, 20 μ m; scanning time, 14 min). The 3D imaging was viewed using the 3D Viewer software of the Quantum GX II. The size and volume of the femur bone were measured via AccuCT analysis software within the ossified diaphysis and metaphysis of the femur, excluding the epiphysis. Bone mineral density (BMD) calibration was performed using a 4.5 mm BMD phantom and BMD measurements were taken at the center of the diaphysis.

Histology

Alcian blue & alizarin red staining was performed on control and SMN2 0-copy *Smn*^{ΔMPC} mice at E18.5 using a previously reported protocol (Ovchinnikov, 2009). For hematoxylin & eosin and toluidine blue staining of the growth plate in the control and SMN2 1-copy *Smn*^{ΔMPC} mice at P14, the femurs were fixed in 4% paraformaldehyde for 24 hours, rinsed in running tap water for 24 hours, and incubated with 10% EDTA (pH 7.4) at 4°C with shaking for 2-3 days. Subsequently, the samples were rinsed in running tap water for 24 hours and then dehydrated through ethanol/xylene and embedded in paraffin. The embedded samples were then sectioned to a thickness of 5 μ m, rehydrated, and stained with Gill No. 3 formula hematoxylin and eosin Y (H&E, Sigma Aldrich, St. Louis, MO, USA) and toluidine blue (Sigma Aldrich, St. Louis, MO, USA). Stained slides were analyzed using 10 $\times$ and 20 $\times$ objectives in the EVOS M7000 imaging system (Thermo Fisher Scientific, Waltham, MA, USA). The growth plate's proliferative and hypertrophic zones were defined by their respective cell sizes.

Hindlimb fibro-adipogenic progenitors isolation

Isolation of limb muscle FAPs was performed according to a previously reported protocol (Kim et al., 2021) with modifications. Limb muscles were dissected and mechanically dissociated in Dulbecco's modified Eagle's medium (DMEM, Hyclone) containing 10% horse serum (Hyclone, Logan, UT, USA), collagenase II (800 units/mL; Worthington, Lakewood, NJ, USA), and dispase (1.1 units/mL; ThermoFisher Scientific, Waltham, MA, USA) at 37°C for 40 min. Digested suspensions were subsequently triturated by sterilized syringes with 20G 1/2 needle (BD Biosciences, Franklin Lakes, NJ, USA) and washed with DMEM to harvest mononuclear cells. Mononuclear cells were stained with corresponding antibodies. All antibodies used in FACS analysis were listed in Supplemental Table 1. To exclude dead cells, 7AAD (Sigma Aldrich; St. Louis, MO, USA) was used. Stained cells were analyzed and 7AAD⁻Lin⁻Vcam⁻Sca1⁺ (Stem cell antigen 1; *Ly6a*) (FAPs) were isolated using FACS Aria III cell sorter (BD Biosciences) with 4-way-purity precision. For western blot, freshly isolated FAPs were cultured at 37°C in alpha-MEM (Hyclone) supplemented with Antibiotic-Antimycotic (anti-anti, Gibco) and 20% FBS (Hyclone). For transplantation, YFP⁺ FAPs and tdTomato⁺ FAPs were isolated from postnatal day 10~21 *Prrx1*^{Cre}; *Rosa26*^{YFP/+} mice and *Prrx1*^{Cre}; *Rosa26*^{tdTomato/+} mice, respectively. Isotype control density plots were used as a reference for positive gating.

Chondrocytes isolation

Isolation of chondrocytes followed a modified protocol previously published (Jonason et al., 2015). Using blunt forceps, cartilage caps were removed from P21 mouse femoral heads and dissected into ~1mm fragments in a petri dish with 10X anti-anti in PBS. The cartilage fragments were washed twice with PBS and then incubated in 5 mL collagenase II solution (800 units/mL collagenase II in DMEM with anti-anti, sterilized by 0.2 μ m filtration) in a 60-mm culture dish at 37 °C in a 5% CO₂ incubator overnight. Chondrocytes were released by pipetting the remaining cartilage fragments ten times and then filtering them through a 70 μ m cell strainer to a 50 ml conical tube. The cells were then washed twice with PBS and pelleted by centrifugation at 500g for 5 min. Subsequently, the cells were cultured overnight at 37 °C in a 5% CO₂ incubator, in complete culture medium (DMEM with 10% FBS and anti-anti) overnight at 37 °C in a 5% CO₂ incubator.

PCR reaction

To detect the presence of *Smn* exon 7 floxed and deleted allele, 20 mg of each tissue were dissolved in direct PCR buffer (VIAGEN) with proteinase K overnight at 65°C. After inactivation at 95°C for 30 minutes, PCR was performed using previously reported primer sets to verify the existence of Cre, *Smn*^{F7}, and *Smn*^{Δ7} alleles (Frugier et al., 2000 [DOI](#)). Primers are listed in Supplemental Table 1.

RNA extraction and measurement of mRNA expression

Total RNA was extracted from the brain, liver, spinal cord L4, tibialis anterior muscle, freshly isolated FAPs, and chondrocytes using a TRIzol Reagent (Life Technologies, Carlsbad, CA, USA) and analyzed by qRT-PCR. First-strand complementary DNA was synthesized from 1 μg of RNA using ReverTra Ace (Toyobo, Osaka, Japan) containing random oligomer according to the manufacturer's instructions. qRT-PCR (Qiagen) was performed with SYBR Green technology (SYBR Premix Ex Taq, Qiagen) using specific primers against indicated genes. Relative mRNA levels were determined using the $2^{-\Delta\Delta C_t}$ method and normalized to *Gapdh* (Figure 1H [DOI](#), Figure 1—figure supplement 1K [DOI](#)). Primers are listed in Supplemental Table 1.

Western blot

Cultured FAPs at passage 3 were homogenized in RIPA buffer (50 mM Tris-HCl, pH 7.5, 0.5% SDS, 20 μg/mL aprotinin, 20 μg/mL leupeptin, 10 μg/mL phenylmethylsulfonyl fluoride, 1 mM sodium orthovanadate, 10 mM sodium pyrophosphate, 10 mM sodium fluoride, and 1 mM dithiothreitol). Cell lysates were centrifuged at 13,000 rpm for 15 min. Supernatants were collected and subjected to immunoblot. BCA protein assay (Thermo Fisher Scientific) was used for estimating total protein concentrations. Normalized total proteins were analyzed by electrophoresis in 10% polyacrylamide gels and transferred to PVDF membranes (Millipore, Billerica, MA, USA). Membranes were blocked in 5% skim milk (BD Biosciences) in TBS with 0.1% Tween-20 and incubated with primary antibodies overnight at 4°C. After incubation with the corresponding HRP-conjugated secondary antibodies, the membranes were developed using a Fusion solo chemiluminescence imaging system (Vilber, Marne-la-Vallée, France). α-tubulin was used as a loading control. Band intensities were quantified using ImageJ software. Antibodies used in this study were listed in Supplemental Table 1. Primary and secondary antibodies were diluted 1:1000 and 1:10000 with PBS containing 0.1% Tween-20 and 3% BSA, respectively.

FAPs transplantation

FAPs transplantation was performed according to a previously reported protocol (Kim et al., 2021) with modifications. YFP⁺ or tdTomato⁺ FAPs (7AAD[−]Lin[−]Vcam[−]Sca1⁺) were isolated by FACS from the limb muscles of indicated mice. 1×10^5 FAPs were suspended in 0.1% gelatin (Sigma Aldrich; St. Louis, MO, USA) in PBS and then transplanted into one side of the TA muscles of SMN2 1-copy *Smn*^{ΔMPC} mice. The contralateral muscle received an equivalent volume of 0.1% gelatin in PBS (Vehicle).

Electrophysiology

The extensor digitorum longus (EDL) muscle was dissected from control and SMN2 1-copy *Smn*^{ΔMPC} mice, along with the peroneal nerve, and then pinned to a Sylgard-coated recording chamber. Intracellular recording was conducted in oxygenated Ringer's solution, which comprised 138.8 mM NaCl, 4 mM KCl, 12 mM NaHCO₃, 1 mM KH₂PO₄, 1 mM MgCl₂ and 2 mM CaCl₂ with a pH of 7.4. Action potential of the muscle was prevented by preparing the muscle in 2.5 μM μ-conotoxin GIIB (Alomone, Jerusalem, Israel) for 10 minutes beforehand. The recording was performed in toxin-free Ringer's solution. Miniature endplate potentials (mEPPs) were recorded from a junction, followed by recordings of evoked endplate potentials (eEPPs) by stimulating the

attached peroneal nerve. The eEPPs were elicited using evoked stimulation. Paired-pulse stimulation (10 ms inter-stimulus interval) was utilized to assess synaptic transmission. The data was obtained and analyzed with Axoclamp 900A and Clampfit version 10.7 software.

Transmission electron microscopy

NMJ transmission electron microscopy followed a modified protocol previously reported (Modla et al., 2010). Mouse EDL muscle was swiftly excised and fixed in 4% paraformaldehyde (PFA) dissolved in Sorensen's phosphate buffer (0.1M, pH 7.2), followed by washing in 0.1 M phosphate buffer. The EDL was then gradually infiltrated on a rotator at room temperature with sucrose: 0.1M phosphate buffer solutions of 30% and 50%, for one hour each, followed by an overnight incubation in 70% sucrose. Excess sucrose was then eliminated using filter paper, and the muscle was embedded in an optimal cutting temperature compound (O.C.T.; Sakura Finetek, Torrance, CA, USA), followed by being frozen in a cryostat (Leica, Wetzlar, Germany). 10- μ m thick longitudinal sections were washed in PBS and treated with Alexa fluor 555-conjugated α -bungarotoxin (1:500, Invitrogen) for an hour. Imaging was conducted with the EVOS M7000 imaging system, and we selected 4-5 NMJ-rich regions for processing with TEM. The sections were fixed with 2% glutaraldehyde and 2% PFA in 0.1M cacodylate buffer (pH 7.2) for two hours at room temperature, with an additional overnight incubation at 4L. After washing with 0.1M cacodylate buffer they were post-fixed with 1% osmium tetroxide in 0.1M cacodylate buffer (pH 7.2)-for two hours at 4°C. The sections were then stained en bloc with 0.5% uranyl acetate overnight, washed with distilled water, and dehydrated using serial ethanol and propylene oxide. The sections were embedded in epoxy resin (Embed-812, Electron Microscopy Sciences) and detached from the slides by dipping them in liquid nitrogen. Ultra-thin sections (70nm) were prepared with a diamond knife on an ultramicrotome (ULTRACUT UC7, Leica) and mounted on 100 mesh copper grids. Sections were stained with 2% uranyl acetate for 10 minutes and lead citrate for 3 minutes, then observed using a transmission electron microscope (80kV, JEM1010, JEOL or 120kV, Talos L120C, FEI). Synaptic vesicle density was quantified within a distance of 500 nm from the presynaptic membrane.

Immunohistochemistry

For NMJ staining, freshly dissected TA muscles were fixed in 4% PFA for 30 min at room temperature. Subsequently, the muscles were cryoprotected in 30% sucrose overnight, embedded in O.C.T., snap-frozen in liquid nitrogen, and stored at -80°C prior to sectioning. Longitudinal 40 μ m-thick sections were obtained from the embedded muscles using a cryostat. The sections were blocked for 2 hours at room temperature using 5% goat serum and 5% bovine serum albumin in PBS/0.4% Triton X-100. Then, the sections were incubated with primary antibodies in the blocking buffer for two days at 4L°C. After washing the sections three times with PBS/0.4% Triton X-100, the sections were stained with secondary antibodies overnight at 4L°C, and then incubated with Alexa fluor 488 or 555-conjugated α -bungarotoxin (1:500, Invitrogen) for 2hr at RT, washed three times with PBS/0.4% Triton X-100 and mounted in VECTASHIELD. Z-serial images were collected at 40X with a Leica SP8 confocal laser scanning microscope. To analyze NMJ morphology, LasX software was used to obtain maximal projections. Neurofilament (NF) varicosity refers to the varicose NF end connected to the rest of the nerve terminal. To quantify NMJ size and synaptophysin coverage, the Btx area and synaptophysin area were measured by ImageJ analysis software.

For bone section staining, the 5 μ m-thick bone sections were rehydrated and antigen retrieval was then performed in citrate buffer (10 mM citric acid, pH 6) at 95°C 20 min. The sections were blocked for 1 hr at room temperature using 5% goat serum and 5% bovine serum albumin in PBS/0.4% Triton X-100. Then, the sections were incubated with primary antibodies in the blocking buffer at 4L°C overnight. After washing the sections three times with PBS/0.1% Triton X-100, the sections were stained with secondary antibodies for 1 hr at RT, washed and mounted. Imaging was conducted with the EVOS M7000 imaging system.

Statistical analysis

All statistical analyses were performed using GraphPad Prism 9 (GraphPad Software). For comparison of significant differences in multiple groups for normally distributed data, statistical analysis was performed by one-way or two-way ANOVA followed by Tukey's pairwise comparison post hoc test. For non-normally distributed data, Brown–Forsythe and Welch ANOVA followed by Games-Howell multiple comparisons test was used. For the comparison of two groups, Student's unpaired t-test assuming a two-tailed distribution with Welch's correction was used. Unless otherwise noted, all error bars represent s.e.m. The number of biological replicates and statistical analyses for each experiment were indicated in the figure legends. Independent experiments were performed at least in triplicates. $P < 0.05$ was considered statistically significant at the 95% confidence level. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

Supplemental information

Supplemental information includes one Supplementary Table.

Ethics

The care and treatment of animals were approved by the IACUC of Seoul National University.

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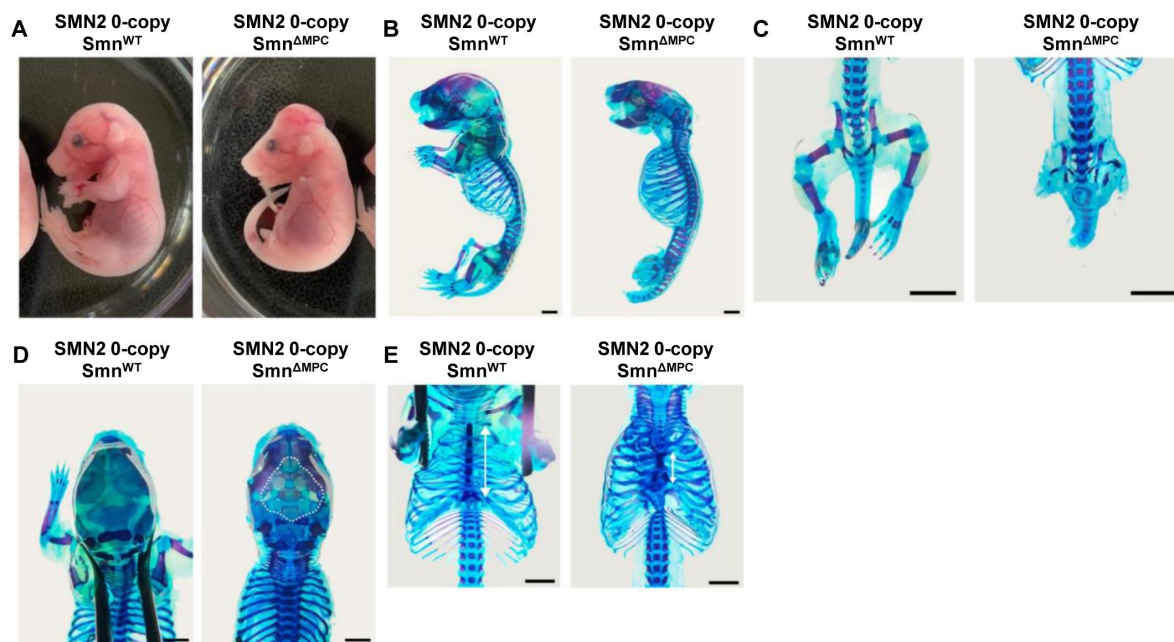


Figure 1 -figure supplement 1

Growth defects in the Prrx1-lineage bone of SMN2 0-copy $Smn^{\Delta MPC}$ mice.

(A) Representative control and mutant mice with 0 copies of SMN2 were photographed at E18.5. (B-C) Alcian blue & alizarine red staining in the SMN2 0-copy $Smn^{\Delta MPC}$ mutant showed abnormal skeletal development of the limbs, (D) calvaria (shown by the dashed line), and (E) sternum (indicated by a white arrow). Scale bars, 5 mm (B-C) and 2 mm (D-E).

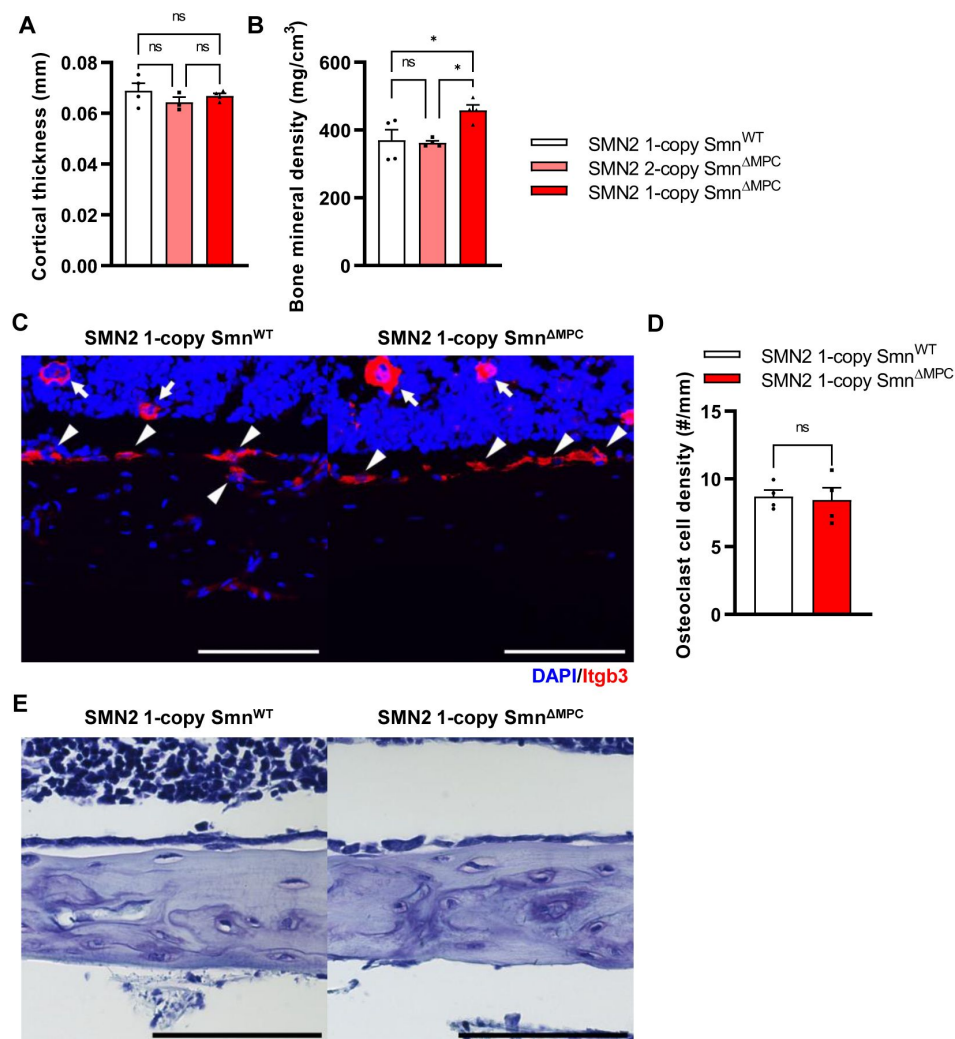


Figure 1 -figure supplement 2

Osteoclasts and osteoblasts were undisturbed in SMN2 1-copy *Smn*^{ΔMPC} mice.

(A) Cortical bone thicknesses were not significantly different between the control and mutant groups. (B) Bone mineral density was slightly increased in the diaphysis of SMN2 1-copy *Smn*^{ΔMPC} mice. The micro CT analysis was performed in femur diaphysis and metaphysis from SMN2 1-copy *Smn*^{WT}, SMN2 2-copy, and 1-copy *Smn*^{ΔMPC} mice at P14. 1-way ANOVA with Tukey's post hoc test, $n = 3-5$ mice in each genotype (A-B). (C) Representative images of osteoclast marker *Itgb3* immunostaining in femur diaphysis cortical bone from mice at P14. Scale bars, 100 μ m. *Itgb3*⁺ hematopoietic cell (indicated by arrow) and osteoclast (indicated by arrowhead) were imaged. (D) It showed similar osteoclast density. $n = 3$ mice in each genotype; Unpaired t-test with Welch's correction. (E) Representative images of toluidine blue staining in femur diaphysis cortical bone for osteoblast evaluation. Scale bars, 100 μ m. ns; not significantly different. $*p < 0.05$. Error bars show s.e.m.

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Reviewer #1 (Public Review):

Summary:

The manuscript by Hann et. al examines the role of survival motor neuron protein (SMN) in lateral plate mesoderm-derived cells using the *Prrx1*Cre to elucidate how changing cell-specific SMN levels coordinate aspects of the spinal muscular atrophy (SMA) pathology. SMN has generally been studied in neuronal cells, and this is one of the first insights into non-neuronal cells that may contribute to SMA disease. The authors generated 3 mouse lines: a *Prrx1;Smn^{f/f}* conditional null mouse, as well as, single and double copy *Prrx1;Smn^{f/f};SMN2* mice carrying either one or two copies of a human SMN2 transgene. First, the bone development and growth of all three were assessed; the conditional null *Smn* mutation was lethal shortly after birth, while the SMN2 2-copy mutant did not exhibit bone growth phenotypes. Meanwhile, single-copy SMN2 mutant mice showed reduced size and shorter limbs with shorter proliferative and hypertrophic chondrocyte zones. The authors suggested that this was cell autonomous by assessing the expression of extrinsic factors known to modulate proliferation/differentiation of growth plate chondrocytes. After assessing bone phenotypes, the authors transitioned to the assessments of neuromuscular junction (NMJ) phenotypes, since there are documented neuromuscular impairments in SMA and the *Prrx1*Cre transgene is expressed in muscle-associated fibro-adipogenic progenitors (FAPs). Neonatal NMJ development was unchanged in mutant mice with two copies of SMN2, but adult single-copy SMN2 mutant mice had abnormal NMJ morphology, altered presynaptic neurotransmission, and problematic nerve terminal structure. Finally, the authors sought to assess the ability to rescue NMJ phenotypes via FAP cell transplantation and showed wild-type FAPs were able to reduce pre/postsynaptic fragmentation and neurofilament varicosities.

Strengths:

The conditional genetic approaches are novel and interestingly demonstrate the potential for chondrocyte and fibro-adipogenic progenitor-specific contributions to the SMA pathology.

The characterizations of the neuromuscular and NMJ phenotypes are relatively strong.

The data strongly suggest a non-neuronal contribution to SMA, which indicates a need for further mechanistic (cellular and molecular) studies to better understand SMA.

Weaknesses:

The skeletal analyses are not rigorous and likely do not get to the core of how SMN regulates bone development.

The overall work is descriptive and lacks convincing mechanisms.

Additional experimentation is likely needed to fully justify the conclusions.

Reviewer #2 (Public Review):

Summary:

Sang-Hyeon et al. laid out a compelling rationale to explore the role of the SMN protein in mesenchymal cells, to determine whether SMN deficiency there could be a pathologic mechanism of SMA. They crossed *Smnf7/f7* mice with *Prrx1Cre* mice to produce *SmnΔMPC* mice where exon 7 was specifically deleted and thus SMN protein was eliminated in limb mesenchymal progenitor cells (MPCs). To demonstrate gene dosage-dependence of phenotypes, *SmnΔMPC* mice were crossed with transgenic mice expressing human SMN2 to produce *SmnΔMPC* mice with different copies of SMN2 (0, 1, or 2). The paper provides genetic evidence that SMN in mesenchymal cells regulates the development of bones and neuromuscular junctions. Genetic data were convincing and revealed novel functions of SMN.

Strengths:

Overall, the paper provided genetic evidence that SMN deficiency in mesenchymal cells caused abnormalities in bones and NMJs, revealing novel functions of SMN and leading to future experiments. As far as genetics is concerned, the data were convincing (except for the rescue experiment, see below); the conclusions are important.

Weaknesses:

The paper seemed to be descriptive in nature and could be improved with more experiments to investigate underlying mechanisms. In addition, some data appeared to be contradicting or difficult to explain. The rescue data were not convincing.

<https://doi.org/10.7554/eLife.92731.2.sa1>**Reviewer #3 (Public Review):**

Summary:

SMN expression in non-neuronal cells, particularly in limb mesenchymal progenitors is essential for the proper growth of chondrocytes and the formation of adult NMJ junctions.

Strengths:

The authors show copy numbers of *smndelta7* in MPC influence NMJ structure.

Weaknesses:

Functional recovery by FAP transplantation is not complete. Mesenchymal progenitors are heterogeneous, and how heterogeneity influences this study is not clear. Part of the main findings to show the importance of SMN expression in non-neuronal cells is partly published by the same group (Kim et al., JCI Insight 2022). In the study, the authors used *Dpp4(+)* cells. The difference between the current study and the previous study is not so clear.

<https://doi.org/10.7554/eLife.92731.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Reviewer #1

Major comments:

1. The authors conclude that the bone growth defects are chondrocyte-specific, highlighting no changes in the IGF pathway. However, other bone cells such as mesenchymal progenitors, osteoblasts, osteocytes, and marrow stromal cells are also lateral plate mesoderm derived and likely have roles in the bone growth phenotypes (a). Additionally, while the size decrease of the proliferative zone was stated, no actual proliferation assays such as BrdU were conducted (b). With the elements being of such small size in the mutants, the defects are likely to be found at the earliest stages of limb development at E11.5-E13.5 and may be due to mesenchymal to chondrocyte transitions or defects in osteoblast lineage development (c). Overall, the skeletal characterization is not rigorous and does not identify even a likely cellular mechanism. Further, a molecular mechanism by which SMN functions in mesenchymal progenitors, chondrocytes, or osteoblast lineage cells has not been assessed (d).

(a, c) As the reviewer commented, it seems to be a very important point to evaluate whether there is any problem in embryonic development from the time of mesenchymal cell condensation of the limb bud to the primary ossification center. However, when Hensel et al evaluated bone growth in P3 of severe SMA mice, the growth defect was not very large, with control femur length 3.5 mm and mutant 3.2 mm. it seems that even if SMN defects occur, there is no major problem with endochondral bone formation in the embryonic period (Hensel et al., 2020).

In this study, the SMN2 1-copy mutant with the bone growth defect was found to have a similar reduction in SMN protein to the severe SMA mouse model in experiments quantifying SMN protein. When Hensel et al. performed an in vitro ossification test on primary osteoblasts from the other severe SMA mouse model (Taiwanese severe SMA), they found no significant difference compared to controls. In femurs at P3 from severe SMA mice, they found no difference in bone voxel density and bone thickness (Hensel et al., 2020). In our data, bone thickness was not different in Figure 1 and Figure 1 – figure supplement 2, and BMD was actually greater. Thus, we believe that osteoblast and osteocyte function does not appear to be impaired by the absence of SMNs. When we looked at cortical osteoblasts in our new Figure 1-figure supplement 2, there did not appear to be a significant difference in density.

Furthermore, it is unlikely that BMSCs contributed to the bone growth we observed up to 2 weeks of age. the *Lepr*+*Cxcl12*+ BMSC population, which constitutes $94\% \pm 4\%$ of CFU-F colonies formed by bone marrow cells (Zhou et al., 2014), is *Prrx1*-positive, and is known to be capable of osteogenesis in vivo, was only shown to differentiate into osteoblasts and form new bone in adults over 8 weeks of age. In the *Lepr-cre*; *tdTomato*; *Col2.3-GFP* mouse model, few cells expressing the osteoblast marker *Col2.3-GFP* are found before 2 months, and only about 3% of femur trabecular and cortical osteocytes express *tdTomato* at 2 months (Zhou et al., 2014). In *Cxcl12-CreER*; *tdTomato*; *Col2.3-GFP* mouse model, the researchers did not find tomato positivity in osteoblasts and osteocytes even after administration of tamoxifen at P3 and analysis 1 year later (Matsushita et al., 2020).

We, therefore, concluded that the bone growth abnormalities observed in SMN2 1-copy mutants are due to problems in endochondral ossification caused by chondrocyte defects and not due to other *Prrx1*-lineage skeletal cells.

(b) According to the reviewer's suggestion, we evaluated cell proliferation in the new Figure 1J-L by performing immunostaining for the Ki67 proliferation marker in growth plates.

(d) As the reviewer pointed out, we enhanced the mechanism study and found the reduction of chondrocyte-derived IGF signaling and hypertrophic marker in new Figure 2. We evaluated the density of osteoblasts and osteoclasts, which can affect bone mineralization. We highlighted the limited impact of BMSCs on bone growth in the first two weeks of life. In a previous study, SMN-deleted osteoblasts did not show any issues with ossification (Hensel et al., 2020). In fact, osteoblast density in the SMN2 1-copy mutant was not different from the control, indicating that the skeletal abnormalities can largely be attributed to deficiencies in endochondral ossification caused by chondrocytes. Since chondrocytes are the local source of IGF and our mutants exhibit phenotypes similar to mouse models with reduced IGF, such as downregulated expression of Igf1 and Igfbp3, downregulated IGF-induced hypertrophic gene expression, reduced AKT phosphorylation, proliferation, and growth plate zone length, SMN-deleted chondrocytes probably showed these phenotypes due to decreased IGF secretion. Now, we added new Figure 2A-C, and E.

1. Is the liver the only organ/tissue that supplied IGF to the chondrocytes or are other lateral plate mesoderm-derived cells potential suppliers? It's not possible to pin SMN deletion in chondrocytes as intrinsic ignoring the other bone cell types that it is depleted from in the Prrx1Cre genetic model.

Recently, Oichi et al. reported that the local IGF source in the growth plate is chondrocytes by in situ hybridization and p-AKT staining (Oichi et al., 2023). When we measured IGF in chondrocytes isolated from articular cartilage, the expressions of Igf1 and Igfbp3 were markedly reduced in chondrocytes with SMN deletion compared to controls (New Figure 2E), suggesting that intrinsic SMN expression in chondrocytes plays an important role in the growth plate.

1. Why is SMN protein being isolated from FAPs to assess levels in the null/SMN2 single copy/double copy mutants when the bone defects are supposed to be a chondrocyte-specific phenotype? This protein expression needs to be confirmed in chondrocytes themselves, and or other Prrx1Cre lineaged skeletal cells.

According to the reviewer's suggestion, we attempted to evaluate the protein levels in chondrocytes of the SMN2 1-copy mutant. However, we were unable to obtain sufficient numbers of chondrocytes, because of poor proliferation of mutant chondrocytes compared to controls in culture conditions. We could obtain $\sim 10^4$ viable cells from 1 mouse of SMN2 1-copy mutant. Therefore, our only options for confirming SMN deletion in chondrocytes were DNA and RNA work. As in the Prrx1-lineage FAPs that the amount of SMN protein correlates with the expression levels of full-length SMN mRNA (Figure 2H-J), we expect that the SMN protein in chondrocytes would be fully depleted due to poor full-length SMN mRNA expression (Figure 2H).

1. Figure 2E should have example images of each type of NMJ characterization.

We revised our figure by adding the example images in new Figure 3E.

1. What are the overall NMJ numbers in the normal formation period? Are these constant into the juvenile period when the authors say the deterioration occurs?

We appreciate the reviewer's constructive comments, and it would be interesting to see if we could see a difference in the total number of NMJs. However, there is one NMJ in every myofiber, and each muscle has hundreds to thousands of myofibers. The technical difficulty of confocal imaging an entire muscle, which can be several millimeters across, precludes experiments that count every NMJ and show a difference. It may be possible to do so by combining clearing and confocal line scanning techniques. In our analysis of the NMJ, the formation of the NMJ in the mutant appears to be normal. Additionally, the number of myofibers seems to be the same, and there may be no difference in the total NMJ number.

1. For transplantation experiments the authors sorted YFP or TOMATO+ cells from the *Prrx1Cre* mice muscles, but refer to them as FAPs. It is known that other cells including tenocyte-like cells, pericytes, and vascular smooth muscle cells are identified by this reporter line. Staining for TOMATO colocalization with PDGFRA would help to clarify this.

In the method 'Hindlimb fibro-adipogenic progenitors isolation' section, we sorted 7AAD-Lin-Vcam-Sca1+ population refers to FAPs. For FAPs transplantation, we also used YFP or TOMATO+ FAPs (7AAD-Lin-Vcam-Sca1+). The 'FAPs transplantation' method section did not specify the FAPs population in detail. This has been fixed in the new method. Sca1 (Ly6a) is an effective marker for identifying FAPs within *Prrx1*-lineage cells, as well as *Pdgfra* (Leinroth et al., 2022).

1. The authors only compare the *SMN2* single copy mutant transplantation to contralateral to show rescue, but how does this compare to overall wt morphology?

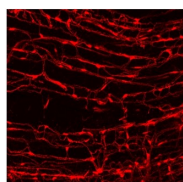
According to the reviewer's constructive comment, we compared them with wild-type morphology (new Figure 7A-D).

1. The asterisks of TOMATO+ in Figure 6A are confusing. FAPs do not usually clump together to form such large plaques and are normally much thinner tendrils. What is the reason for this?

As the reviewer states, FAPs have a fibroblast-like morphology with elongated thinner tendrils. The Figure 6A image in the figure shows a Z-sliced cell body portion of FAP, where the nucleus is located, and it appears blunt. We attached imaged tomato+ FAPs, in which their cell body parts are plaque-like.

Author response image 1.

Tomato+ FAPs in muscle



1. Would transplantation of healthy FAPs after NMJ maturation in *SMN* mutants still rescue the phenotype? Assessment of this is key for therapy intervention timelines moving forward.

It will be very interesting to see if the phenotype improves after NMJ maturation by healthy FAPs transplantation, but this is a technically difficult experiment to do because we found that FAPs do not implant effectively when injected into naive adult muscle. The transplantation into the adult is sufficiently possible if accompanied by an injury, but this eventually leads to new formation of NMJ again. Thus, it seems impossible to do transplantation experiment after NMJ maturation through general methods. If we discover a method to efficiently rescue SMNs from FAPs or identify a factor that affects FAPs' influence on NMJ, then we may be able to conduct this experiment.

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Reviewer #2

Major comments:

1. Regarding bone deficits - CT analysis of bones should be more comprehensive than Figure 1A shows. How about cross-sections? (a) Are bone phenotypes also age-dependent? (b) PCR was done only for SMA and related proteins (such as IGF). IGF protein in the blood and relevant organs should be studied. Why not include biomarkers of osteoblasts or/and osteoclasts and their regulators? (c)

(a) We appreciate the reviewer's constructive comment. we added longitudinal section views in new Figure 1A and a description of trabecular bone volume and secondary ossification center in the main text.

(b) Age-dependent evaluation is an important point. By adulthood, the difference between the SMN2 1-copy mutant and the control is much larger, and even at birth there is a slight difference, although not as large as at 2 weeks of age. We focused our phenotyping on bone growth at 2 weeks of age, a time when new bone formation by BMSCs is less influential, when

bone growth is primarily driven by endochondral ossification of chondrocytes, and before the defect in the NMJ is primarily manifested.

(c) As the reviewer comments, it is important that IGF are evaluated in tissues other than liver. However, the liver is most likely the source of systemic IGF, as shown by the liver-specific deletion of *Igf1* and knockout of *Igfals*, a protein that forms the IGF ternary complex, which is predominantly expressed in the liver. This resulted in a 90% drop in serum IGF levels and a phenotype of shortened femur length and growth plates in the double KO mice (Yakar et al., 2002).

The local IGF source in the growth plate is chondrocytes confirmed by *Igf1* in situ hybridization and p-AKT staining (Oichi et al., 2023). From the In situ hybridization data, we can observe that bone marrow and bone do not express *Igf1* at all, but only perichondrium and chondrocytes in the resting zone express *Igf1* mRNA. Therefore, we can see that the only supplier of IGF among LPM-derived cells is chondrocytes, and in the new figure 2, we measured IGF pathway expression and AKT phosphorylation in chondrocytes. We have confirmed that the expression of *Igf1/Igfbp3* is reduced in chondrocytes with *SMN* deletion.

To assess serum IGF level, we could not set up this experiment condition during our revision period due to the requirement of administrative procedures for purchasing new apparatuses and the limitation of our research funds. However, as previously stated, there is no difference in the expression of *Igf1* and *Igfals* in the liver, which accounts for 90% of serum IGF levels. Therefore, we did not anticipate significant variations in serum IGF levels.

Evaluation of osteoblasts or osteoclasts was done by section staining due to sampling difficulties for PCR. we assessed osteoblasts and osteoclasts state in new Figure 1-figure supplement 2.

1. *What is the relationship between deficits of bone deficits and muscle deficits or even NMJ deficits? Are they inter-related? Is skeletal muscle development also defective in *SmnΔMPC* mice? Can NMJ deficits result from bone deficits? Or vice versa?*

Unfortunately, the reviewer's comments are very difficult to clarify in our study using the *Prrx1-cre* model. In skeletal muscle development, the myofiber number was not significantly different in our mouse models. A study has shown that inactivating *noggin*, a BMP antagonist expressed in condensed cartilage and immature chondrocytes, results in severe skeletal defects without affecting the early stages of muscle differentiation (Tylzanowski et al., 2006). Therefore, bone may not have a significant impact on the early development of muscle, but later in postnatal development it may have an impact on motor performance issues. The relationship between bone and NMJ hasn't been studied. The impact of bone defects on motor skill may result in muscle weakness and NMJ problems. In our study, we showed that NMJ deficit rescue by transplantation of FAPs and decreased IGF in chondrocytes, a key source of local IGF. This suggests that the functions of FAPs in NMJ and chondrocytes in bone deficit are crucial, rather than each other's influence.

1. *Regarding the rescue experiment, the interpretation of the data should be careful. Evidently, healthy FAPs (td-Tomato positive) were transplanted into TA muscles of 10 days-old *SMN2* 1-copy *SmnΔMPC* mice, and NMJs were looked at P56. The control was contralateral TA that was injected with the vehicle. As described above, the data had huge SEM and were difficult to interpret or believe. The control perhaps was wrong if FAPs act by releasing "chemicals" because FAPs from one leg may go to other muscles via blood. Second, if FAPs act via contact, the data shown did not support this. Two red FAPs were shown in Figure 6, one of which was superimposed with a nerve track to one of the three NMJs. This NMJ however did not show any difference to the other two, which did not support a contact mechanism. These rescue data were not convincing.*

We appreciate the reviewer's critical comment, but the reviewer appears to have confused the minimum and maximum range bars in the box-and-whisker plot with the SEM error bar in the bar graph. We apologize for the insufficient description of the figure legends section. We revised them. New Figure 7C, which is a bar graph, has a sufficiently short SEM error bar. In contrast, box-and-whisker plots B and D depict the minimum and maximum range, instead of the SEM, and they are significantly different with a p-value of less than 0.001. If FAPs affect the NMJ via a paracrine factor or ECM with a short range of action, they may rescue the NMJ defect in a non-contact-dependent manner, without affecting the contralateral muscle. Also, the FAPs are heterogeneous, so if only a certain subpopulation rescues, the tomato+ FAP in the figure may not be the rescuing cells.

1. For most experiments, the "n" numbers were too small. 3-5 mice were used for bone characterization. For the NMJ, most experiments were done with 3 mice. It was unclear how many NMJs were looked at. Perhaps due to small n numbers, the SEM values were enormous (for example, in Figure 6).

As with the response to the previous comment, this is due to confusion between box-and-whisker plots and bar graphs, and our data was determined to be significant using the appropriate statistical method.

1. Also for experimental design, some experiments included four genotypes of mice (Fig. 1 J,K) whereas some had only three (Fig. 1 A, B, C, D and Fig. 3) and others had two (many other figures).

In the first experiments to confirm the phenotypes, we tested the 2-copy mutant, but it was not significantly different from the wild type, and in subsequent experiments, we mainly tested the only 1-copy mutant.

1. What was the reason why mixed muscles were used for NMJ characterization (TA versus EDL)? Why not pick a type I-fiber muscle and a type II-fiber muscle?

We appreciate the constructive comment from the reviewer. Firstly, we conducted a phenotype analysis on the TA muscle. For electrophysiological recording, the EDL muscle should be used for intact nerve with muscle preparation, technically. Additionally, for TEM imaging, EDL was a suitable muscle to locate NMJ positions before TEM processing. Both TA and EDL muscles are adjacent and have similar fiber-type compositions. It would be important to observe in different fiber types of muscles, but when we first identified the phenotype, various types of limb muscles showed similar defects, so we focused on specific muscles.

1. The description of mouse strains was confusing. SMN2 transgenic mice (with different copies) were not described in the methods.

We apologize for the insufficient description of the method section. By crossing mice with the SMN2^{+/+} homologous allele, SMN2 heterologous mice with only one SMN2 allele are SMN2 1-copy mice (SMN2^{+/-}) and SMN2 homologous mice are SMN2 2-copy mice (SMN2^{+/+}). We revised our manuscript method 'Animals' section.

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Reviewer #3

1. The authors used *Prrx1Cre* mouse with floxed *Smn* exon7(*Smnf7*) mouse carrying multiple (one or two) copies of the human *SMN2* gene. Is it expressed both in chondrocytes and mesenchymal progenitors in the limb?

We appreciate the reviewer's comment. We analyzed the deletion of *Smn* in chondrocytes and FAPs via Cre using genomic PCR and qRT-PCR, as depicted in new Figure 2. The *SMN2* allele, which is expressed throughout the body, can rescue *Smn* knockout mouse lethality (Monani et al., 2000). Indeed, the short limb length and lethality observed in *SMN2* 0-copy mutants were mitigated by the presence of multiple copies of *SMN2*. Therefore, both Chondrocytes and FAPs may express *SMN2* transcripts from the transgenic *SMN2* allele.

1. Page 10 regarding Fig.2E, please show pretzel-like structure. In Figure 2E, plaque, perforated, open, and branched are shown; however, the pretzel is not shown. The same issue is for the Fig. 3D explanation in the text on page 12.

We appreciate the reviewer's constructive feedback. We included illustrative figures of all types of NMJ characterization, and the branched type is identical to the pretzel type. Therefore, we have replaced 'branched' with 'pretzel' in our text and revised Figure 3E by incorporating the example images.

1. The explanation of the electrophysiology for Fig.4 in the text on pages 12 and 15 (RRP) is not so convincing for the readers. It is advisable to add TEM data for transplantation if it is not technically difficult.

We appreciate the reviewer's critical feedback. Because we did not measure RRP directly, we removed speculation about the possibility of RRP difference. If observing the active zone with TEM and the docking synaptic vesicle would help quantify RRP, it is technically difficult to obtain images of sufficient quality to distinguish the active zones with our current TEM imaging technique.

1. The authors used the word FAP for *7AAD(-)Lin(-)Vcam(-)Sca1(+)*. It is recommended to show the expression of *PDGFR alpha*. Furthermore, as the authors stated in the text, mesenchymal progenitors (FAPs) are heterogeneous. Please discuss this point further. Other reports show at least 6 subpopulations using single-cell analyses (Cell Rep. 2022).

In the report, Ly6a (*Sca1*) is a good marker for FAPs, as well as *Pdgfra* (Leinroth et al., 2022). The 6 subpopulations expressed Ly6a. The one of subpopulations associated with NMJ was discovered. This population expressed *Hsd11b1*, *Gfra1*, and *Ret* and is located adjacent to the NMJ and responds to denervation, indicating an increased possibility of interaction with the NMJ organization. In further our study, we aim to determine which subpopulations are crucial for NMJ maturation by transplanting them to mutants for rescue.

| 1. How do authors determine the number of FAP cells for transplantation?

The FAPs transplantation was performed according to a previously reported our study (Kim et al., 2021).

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